

NEAR-INFRARED CLASSIFICATION OF VARIABLE STARS IN THE VVV SURVEY

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1. Introduction

The VISTA Variables in the Via Lactea (VVV) Survey

A near-infrared (NIR) large-scale survey designed to explore the Galactic bulge and adjacent southern disk, capturing over 288 million sources (Ferreira Lopes et al. 2020).

The Telescope

- 4.1-meter VISTA Telescope
- Located at ESO's Paranal Observatory, Chile.
- High spatial resolution (0.34 arcsec/pixel) to mitigate extreme crowding in dense fields.

VIRCAM & NIR Bands

- Equipped with the 16-detector VIRCAM.
- Operates across 5 NIR bands:
Z (0.87 μm), Y (1.02 μm)
J (1.25 μm), H (1.65 μm)
K_s (2.14 μm)

Data & Strategy

- Spans 562 square degrees across 348 observation fields.
- High-cadence time-series: ~70 epochs over 6 years (2010–2016).
- Photometric precision < 0.05 mag for objects brighter than K_s ~ 16 mag.

Reddening & Extinction

Scattering Regimes

$$x = \frac{2\pi a}{\lambda}$$

- **Optical ($\lambda < 0.8 \mu\text{m}$):** Dominated by Mie Scattering ($x \sim 1$). High interaction with Fe/Si-rich grains.
- **Infrared ($\lambda > 0.8 \mu\text{m}$):** Follows a Rayleigh-like power law ($A_\lambda / A_V \propto \lambda^{-1.7}$) due to larger grains ($a \sim 0.1\text{-}0.3 \mu\text{m}$) (Li et al. 2024).

Reddening Effect: Shorter wavelengths suffer stronger attenuation, shifting the Spectral Energy Distribution (SED) towards the red.

Why K_s Band?

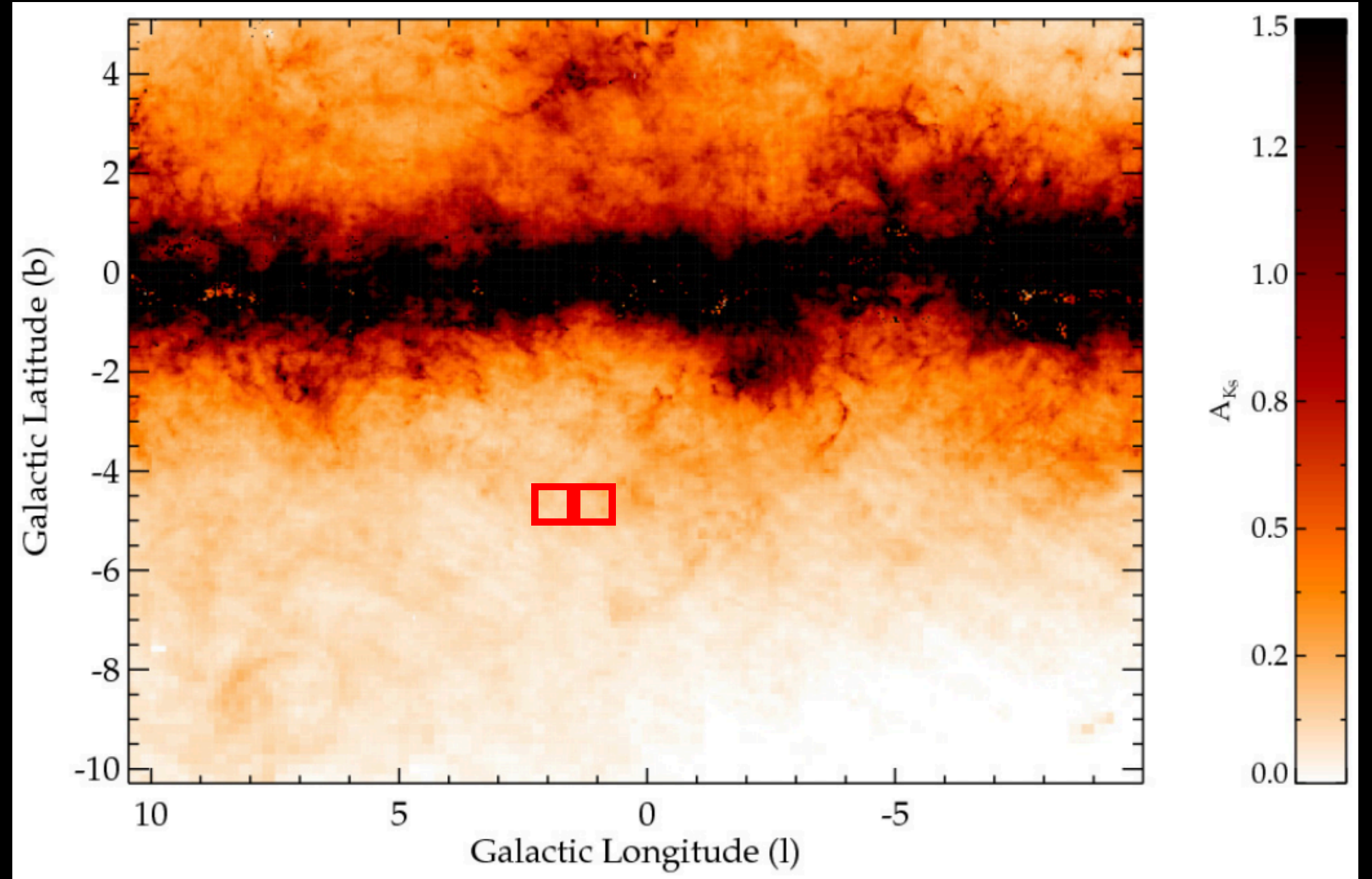
Near-Infrared (NIR) observations achieve 10x deeper penetration into dusty environments than optical surveys (Cardelli et al. 1989).

$$\frac{A_{K_s}}{A_V} = 0.112 \pm 0.002$$

In practical terms: K_s -band measurements suffer only ~10% of the extinction that affects visual wavelengths.

Target Fields: b278 & b279

- **Location:** Inner Galactic Bulge.
b278: $(l, b) \sim (1.09^\circ, -4.23^\circ)$
b279: $(l, b) \sim (2.57^\circ, -4.23^\circ)$
- **Extinction:** Transition zone between the heavily obscured plane ($|b| < 2^\circ$) and clearer mid-bulge regions.
- **Density:** $\sim 10^5$ stars deg^{-2} .



Moderate Extinction: $A_{Ks} \sim 0.32$ mag. Taken from Gonzalez et al. (2011)

Stellar Populations: The Variable Universe

The Galactic Bulge hosts a dense mix of both **constant** and **variable** stars.

Radial Pulsators (Intrinsic)

Stars that physically expand and contract, oscillating around hydrostatic equilibrium (like a "breathing" star).

The Pulsation Engine:

- κ -mechanism: Opacity changes in partially ionized layers trap radiation, causing the envelope to expand.
- γ -mechanism: Variations in energy transport efficiency as layers compress and expand.

Key Targets in this Study

- **Cepheids**

Massive, evolved stars. Fundamental luminosity calibrators via the Period-Luminosity relation.

- **RR Lyræ**

Old, low-mass horizontal branch stars. Excellent tracers of ancient Galactic populations.

Stellar Populations: The Variable Universe

Eclipsing Binaries (Extrinsic)

Systems where variability is purely geometric, caused by mutual eclipses as stars orbit one another.

- **Algol (EA)**: Detached systems with very distinct primary eclipses.
- **β Lyræ (EB)**: Close binaries gravitationally distorted into ellipsoidal shapes.
- **W Ursæ Majoris (EW)**: Extremely compact systems that share a common stellar envelope.

Why are they important?

Identifying and classifying these specific variables is crucial for:

- Chemical Composition: Probing metallicity gradients across the Galactic Bulge.
- Stellar Ages: Dating ancient vs. recent star formation epochs.
- Galactic Evolution: Mapping the structural and kinematic history of the Milky Way.
- Distance Scales: Calibrating distances using standard candles and binary geometries.

Minimum Entropy Vs Traditional Methods

Traditional Methods (e.g., Lomb-Scargle)

Fourier-based methods infer periodicity by fitting sinusoidal functions to the data.

- **Shape Bias:** They assume harmonic signals. Performance severely degrades with non-sinusoidal variability (like eclipsing binaries or RR Lyrae).
- **Aliasing:** Highly vulnerable to pseudo-aliasing and spectral leakage caused by regular observing cadences (e.g., diurnal sampling).
- **Computational Cost:** Often require $O(N \log N)$ operations.

The Entropy Advantage

Minimum Entropy evaluates how well a trial period organizes the data into a coherent phase distribution.

- **No Shape Assumptions:** Ideal for complex, non-sinusoidal light-curve morphologies.
- **Highly Robust:** Naturally handles the sparse, irregular sampling and extreme noise typical of Galactic Bulge surveys (VVV).
- **Efficiency:** Relies on histogramming phased data, a linear-time operation, allowing it to scale to millions of sources.

Minimum Entropy Method: Period Detection Framework

Core Concept: Folding a time series with its true period maximizes phase-space order (Minimum Entropy). Incorrect periods scatter points randomly (High Entropy).

1. Phase Folding

Map observations (t_i, m_i) into a normalized phase ϕ_i using a trial period p .

2. Grid Partitioning

Divide the phase-magnitude plane into $m \times m$ cells. Calculate the probability $\mu_k = n_k / N$ of finding points in each cell.

3. Minimize Shannon Entropy

Compute the entropy. The period T that yields the lowest $S(p)$ is the true variability period.

$$\phi = \left(\frac{t - t_0}{p} \right) \text{mod} (1)$$

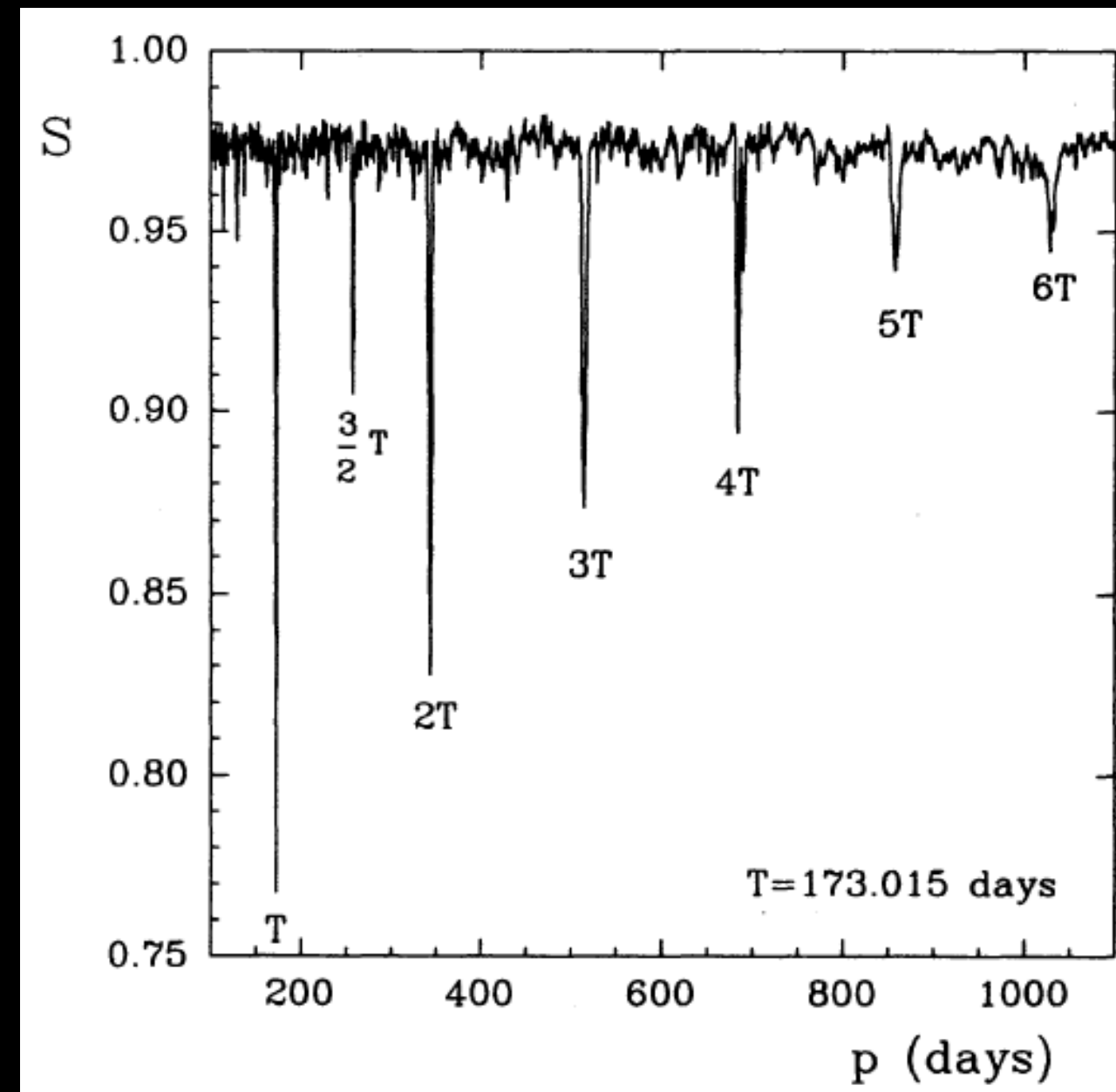
$$S(p) = - \sum_{k=1}^m \mu_k \ln(\mu_k)$$

Machine Learning Integration

Unlike classical methods, this one allows us to extract physically interpretable features.

Minimum Entropy Method: Period Detection Framework

Core Concept: Folding a time series with its true period maximizes phase-space order (Minimum Entropy). Incorrect periods scatter points randomly (High Entropy).



Entropy periodogram of a synthetic star. Taken from Cincotta et al. (1995)

Main Research Objective

To develop a robust integrated **machine learning pipeline** for stellar **variability classification**, applied to fields b278 and b279 of the VVV Survey.

Methodology

Implementing the Minimum Entropy method as a physically interpretable foundation for period detection.

Target

Automated identification and classification of variable stars in high-extinction Galactic Bulge regions.

2. Data Pre-Processing

Data Compilation & Outlier Treatment

- **Origin:** VISTA Science Archive (VSA) - VVV Survey.
- **Target Tiles:** b278 and b279.
- **Criteria:** $\sigma_{Ks} < 0.2$ mag and minimum 30 epochs (Smith et al. 2025).

Quality Filters

- Discarded ambiguous_match = 1 to eliminate blending and crowding effects in dense regions.
- Hold time series with at least 50 epochs.

3 σ Deviation Clipping

✓ **Sparse Sampling:** High performance for stars with ~ 50 epochs.

✗ **Dense Sampling:** Fails to remove high-density clusters of outliers below the 3 σ threshold.

Assumption: Gaussian distribution of magnitude errors.

98% Percentile Retention

✓ **Consistency:** Effectively mitigates low-deviation outliers in high-density light curves.

✗ **Limited Data:** Risks removing true signal in stars with < 60 measurements.

Strategy: Retain points closest to the stellar mean magnitude.

Saturation & Faint-end Thresholds

The Magnitude Challenge:

Defining limits is non-trivial; literature shows significant variance for VVV data:

Author	Ks Limit
Andrews et al. (2014)	> 12.2 mag
Dékány et al. (2015)	11.0 - 15.5 mag
Meingast et al. (2016)	> 11.0 mag
Duplancic et al. (2024)	10.0 - 16.2 mag
VIRCAM/VISTA Manual (2009)	12.0 - 14.0 mag

Variations due to local extinction, crowding, and seeing conditions.

Proposed Methodology

- Saturation (Bright): Defined by the inflection point of the linear Error–Magnitude relationship.
- Faint Limit: Established at a 90% completeness level to discard high-noise sources.

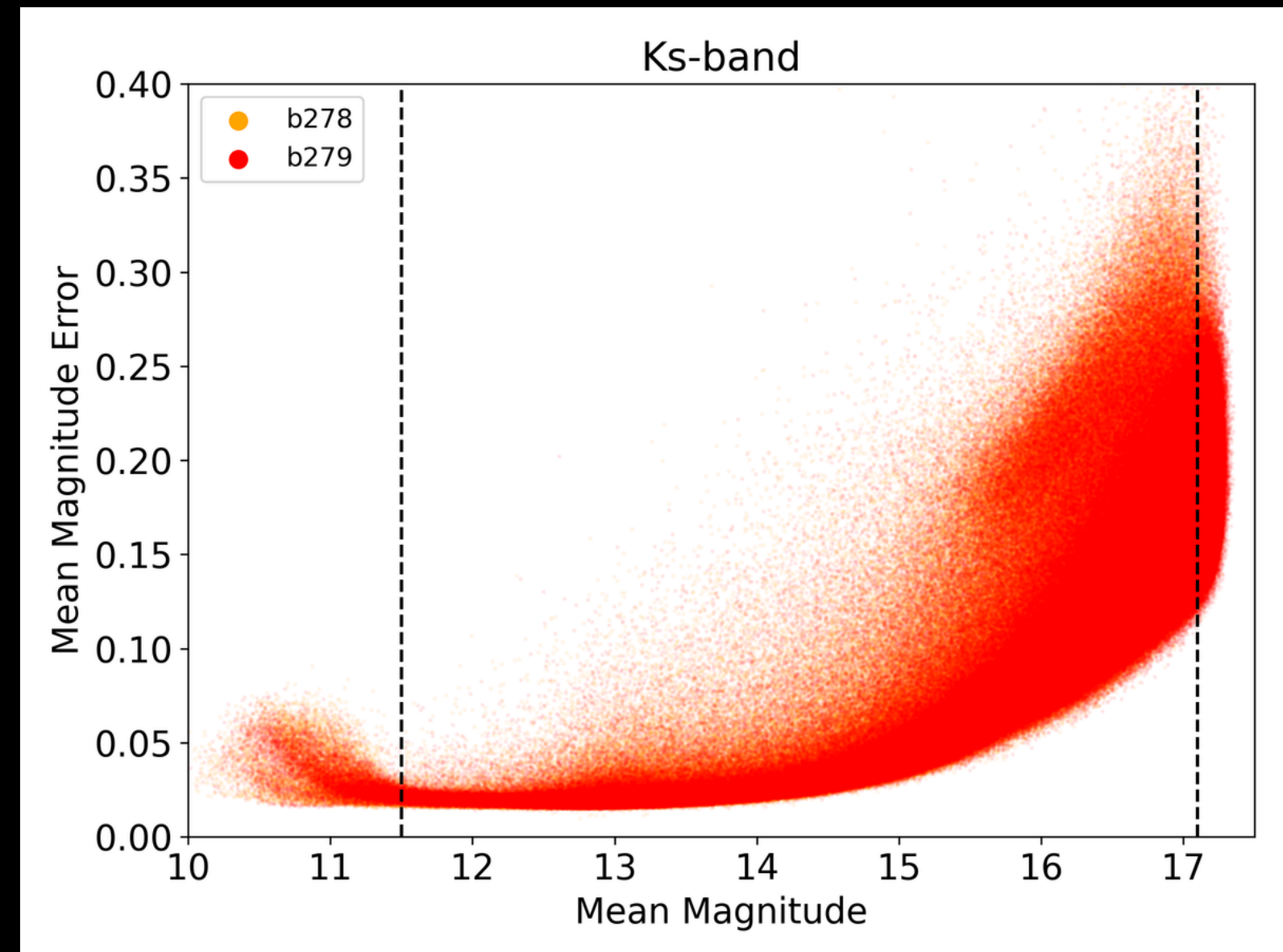
Saturation & Faint-end Thresholds

Final *Nice* Ranges

- b278: [11.5, 17.09] mag
- b279: [11.5, 17.12] mag

Star Counts

	Field b278	Field b279
Nice	1.554.449	1.306.117
Saturated	3.061	3.051
Faint	179.314	146.249



3. Binary Classification Framework

Visual Variability Classification: Building the Ground Truth

The Challenge

There is currently no definitive catalogue of variable and non-variable stars within the VVV project that could be directly used to train our classifiers.

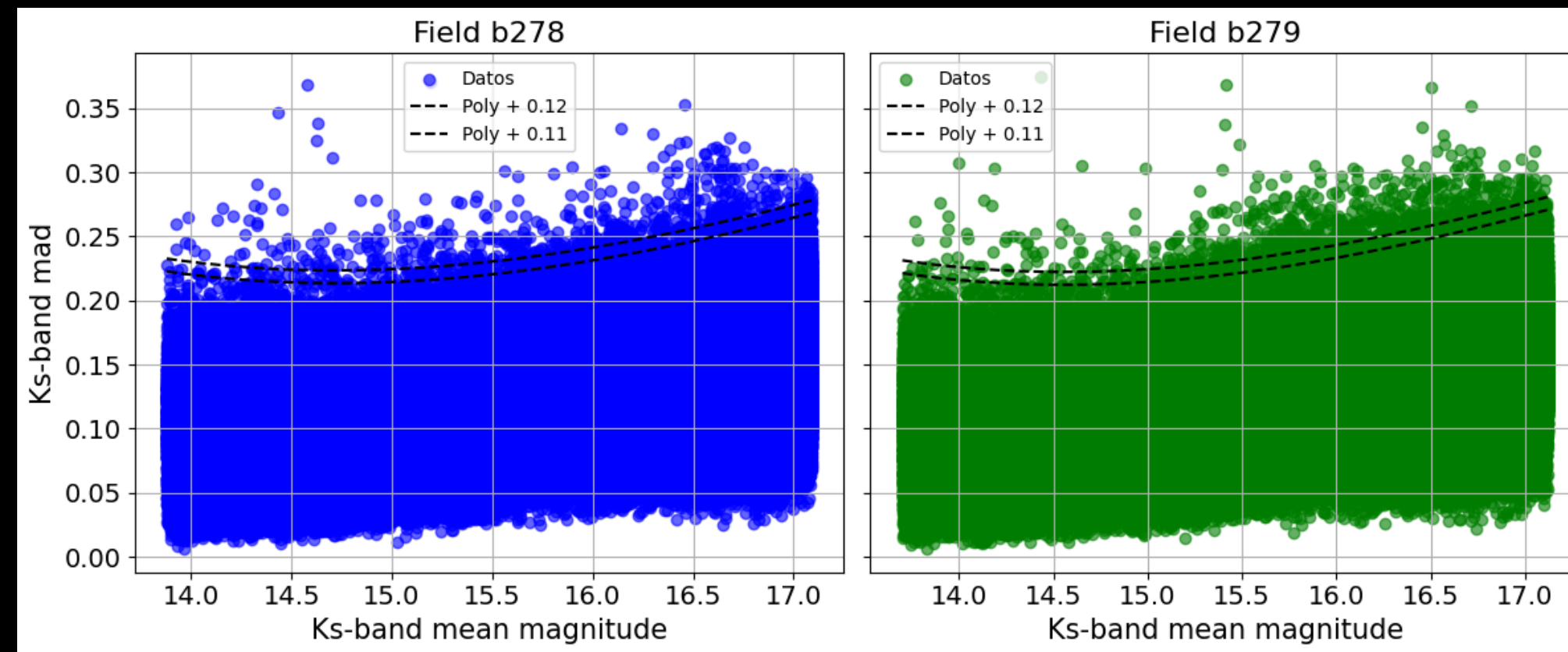
Consequently, we constructed our own manually labeled dataset by visually identifying sources from the previously filtered nice dataset.

MAD vs. Mean Magnitude

To systematically sweep the dataset, we analyzed the Median Absolute Deviation (MAD) against the Mean Magnitude.

- Fitted a 3rd-order polynomial, shifted to create selection bands.
- Expectation: Variable stars (VAR) populate high-MAD regions.
- Expectation: Non-variable stars (NV) span across all MAD ranges (from precise constant sources to heavily noisy artifacts).

Visual Variability Classification: Building the Ground Truth



To prevent selection bias against variables that do not perform well with entropy, we implemented a complementary approach.

1. Shannon Entropy

Evaluate grids (0.01-1.2d & 1.2-95.5d).
Phase-fold best periods.

Coherent → VAR

Diffuse/Scattered → Proceed



2. Lomb-Scargle (LSP)

Re-examine diffuse sources computing
the power spectrum frequency.

Coherent → VAR

Spurious → NV

Ground Truth Dataset Statistics

19,194

Total Stars Inspected

287

Confirmed Variable (VAR)

1,008

Confirmed Non-Variable (NV)

Statistical Backing (Chernoff Bounds)

- The resulting number of labeled VAR and NV sources ensures that the data partition used for training and validation is statistically representative.
- Guarantees a 95% confidence level with a 5% error margin in the estimation of model performance metrics.
- Validates the robustness of metrics such as Accuracy, Precision, and F1-Score across both fields.

Natural Population Asymmetry.

- Only ~1.5% of randomly inspected stars were variables. Reflects the true Galactic Bulge population.
- High photometric noise at ~17 mag limits variable detection.
- We intentionally captured complex NV morphologies (e.g., blending, artifacts) to teach the models to separate them from true variables.

Model Inputs: Classification Features

To translate light curve morphology into machine learning inputs, we extracted physically interpretable features, standardized to optimize training.

Model Inputs: Classification Features

Mean Entropy ($\bar{S}$)

Average uncertainty of the information across trial periods. High values suggest intrinsic variability; low values indicate purely random noise.

Cumulative ΔS

Sum of the prominence of the 20 deepest entropy minima. Captures the depth of the primary periods and their harmonics.

Variability Index (χ^2_{var})

Evaluates the magnitude variance weighted by instrumental photometric errors. Values $\gg 1$ indicate genuine physical fluctuations.

PEW

Normalizes the depth of entropy minima relative to photometric noise. Resolves noise-induced ambiguities.

Model Inputs: Classification Features

Sinusoidal Fits

Measures the amplitude (A_{sin}) and goodness of fit (χ^2_{sin}) of the phased light curve against a pure sinusoidal model.

VPEW

A 40-element sequence that tracks the entropy decay profile. Designed for pattern extraction in a 1D-CNN.

Stab Index (Stab)

Measures the stability under synthetic noise contamination. Crucial for distinguishing between coherent (periodic) changes and stochastic transients.

Statistical Analysis: Feature Validation

Non-Normal Distributions

Shapiro–Wilk tests confirmed non-Gaussian structures.

Heteroscedasticity (Levene)

Distinct and separable variances were confirmed between the VAR and NV classes.

Independence (Spearman Rank)

Minimal redundancy; each feature contributes unique information to the classifier.

Statistical Discriminability

Given that the variables are non-normal and heteroscedastic, we applied the non-parametric Kruskal–Wallis test.

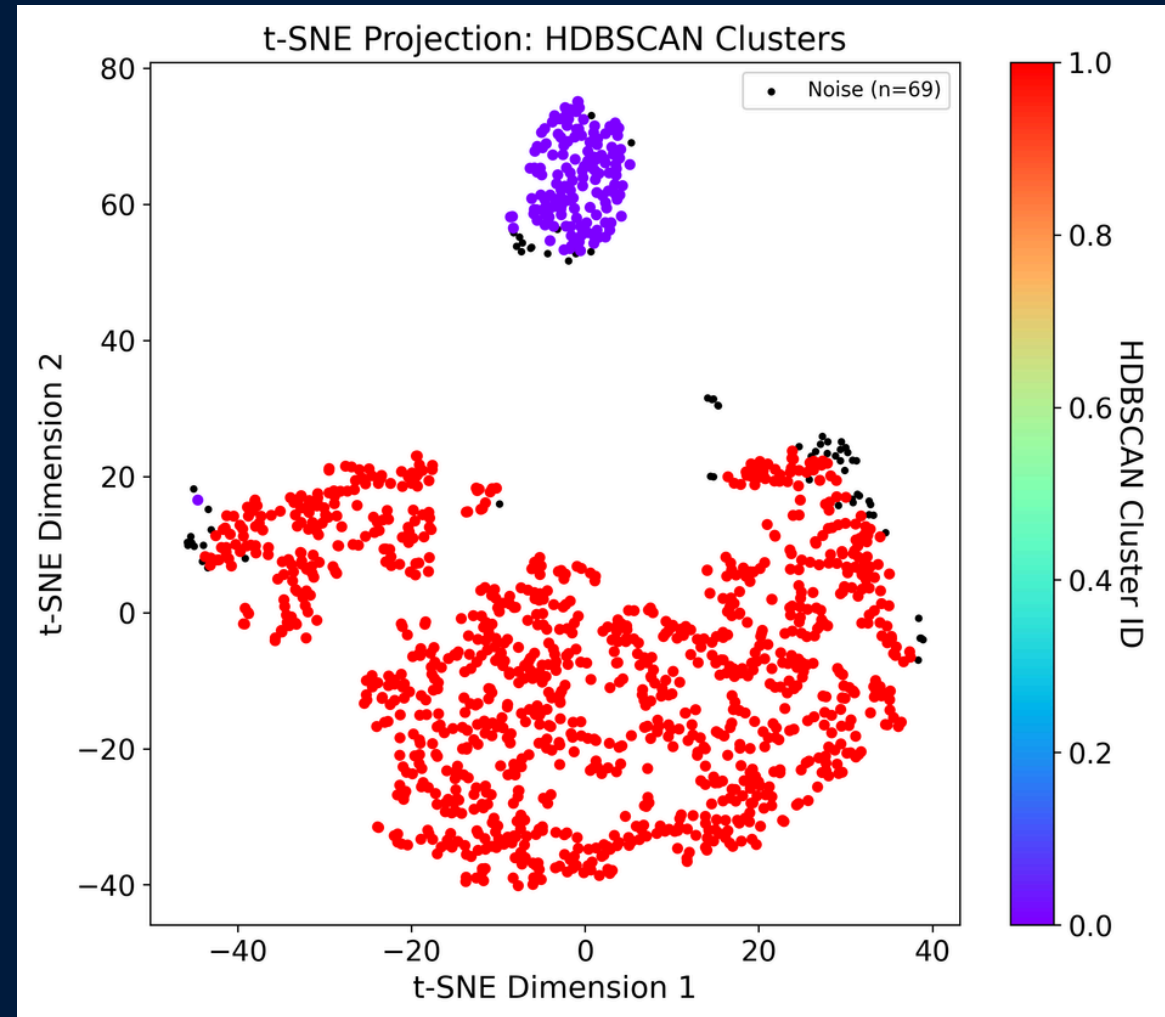
$p\text{-value} < 0.05$ for all variables.

Each extracted feature is statistically significant for distinguishing between the VAR and NV classes.

Binary Predictive Models

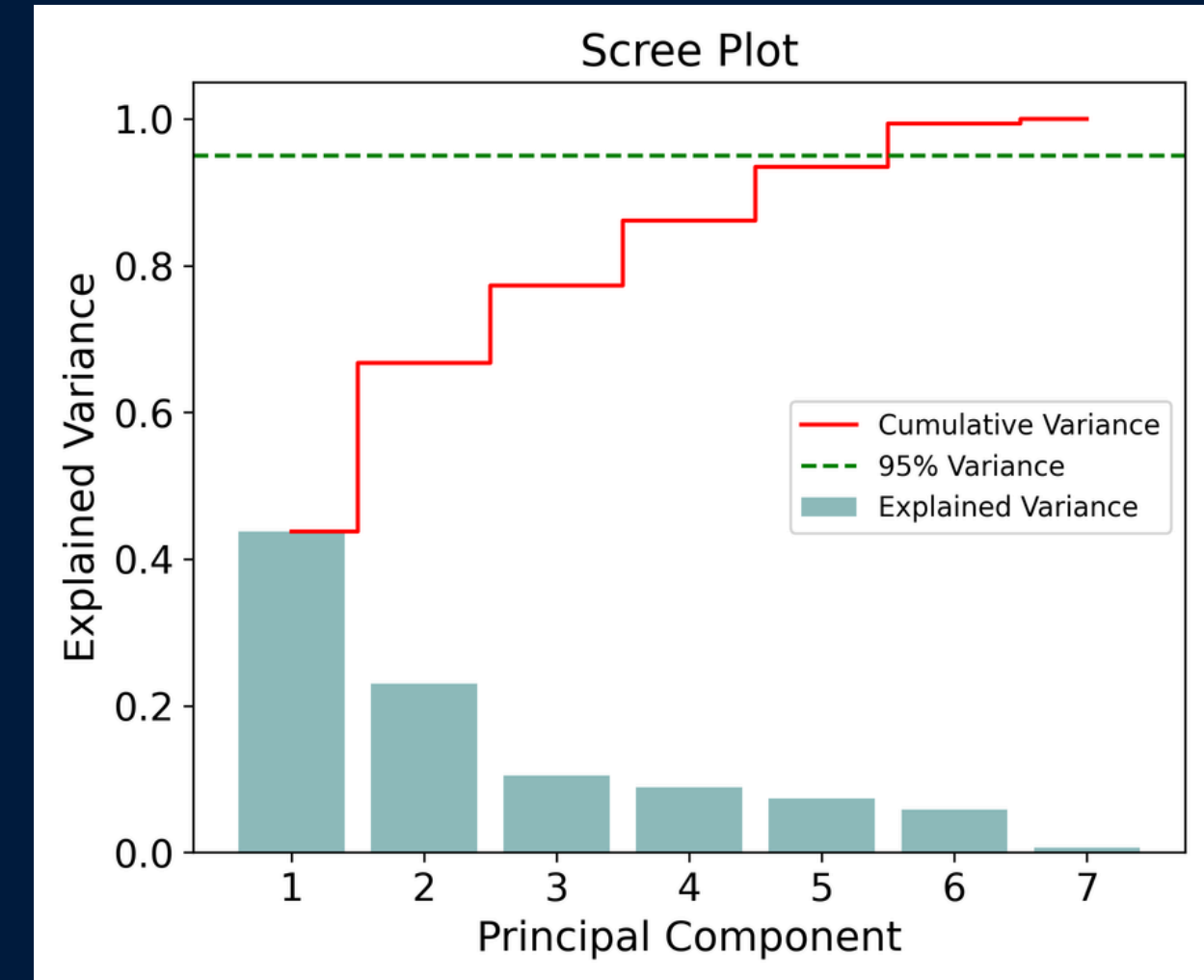
Unsupervised Exploration: HDBSCAN & PCA

Class Separation (t-SNE)



- HDBSCAN effectively separated the VAR and NV classes without prior labels, yielding highly pure clusters (Cluster 0: 99.3% VAR; Cluster 1: 88.8% NV).
- Revealed inherent substructure: The NV class split into two main sub-clusters corresponding to high vs. low photometric dispersion.

Feature Relevance (PCA)



- Principal Component Analysis (PCA) confirmed that all extracted features are relevant.
- The Scree plot demonstrates that at least 6 features are required to describe the 95% of the total data variance.

Supervised Models: Training Strategies

We deployed two distinct architectures, **Random Forest (RF)** for scalar features and a **1D-CNN** for the sequential VPEW array, trained under strict ML protocols to combat the extreme class imbalance.

Hyperparameter Optimization

Extensive grid searches were performed to tune both models, preventing overfitting while maximizing sensitivity toward the minority variable class.

Balanced Class Weights

To counteract the natural 1.5% (VAR) vs 98.5% (NV) distribution, models were penalized heavily for misclassifying the minority VAR class.

Bootstrapping

Utilized random sampling with replacement to ensure the models are exposed to diverse morphological variations, increasing generalization.

4-Fold Cross Validation

Stratified k-fold validation ensures that the performance metrics are robust, consistent, and independent of any specific data split.

The Joint Bayesian Classifier

Probability Histograms to PDFs

- Extracted the raw VAR-class probability outputs (P_{RF} , P_{CNN}) from both independent models.
- Utilized these histograms to estimate continuous Probability Density Functions (PDFs) for the VAR and NV populations.

The Logit Transformation

- Raw probabilities are bounded $[0, 1]$, making Gaussian/KDE fitting mathematically unstable at the edges.
- Applied the logit adjustment: $\text{logit}(p) = \ln[p/(1-p)]$
- This maps the probabilities into an unbounded continuous space $(-\infty, \infty)$, allowing for robust, smooth PDF estimation.

Posterior Probability Integration

The final classification relies on Bayes' Theorem, integrating the evidence (likelihood ratios Λ) from both models:

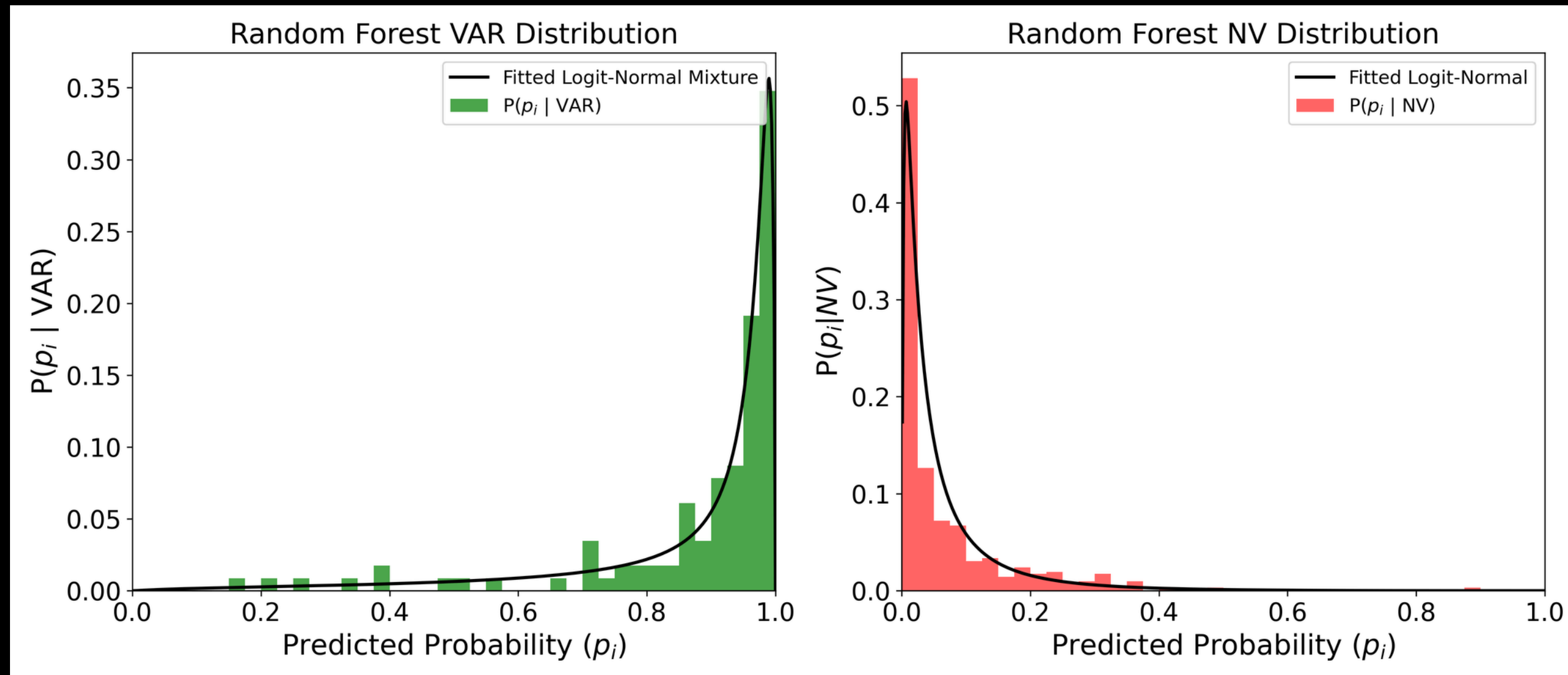
$$P(\text{var}|p_{rf}, p_{cnn}) = \frac{P(p_{rf}, p_{cnn}|\text{var}) \cdot P(\text{var})}{P(p_{rf}, p_{cnn})}$$

$$P(\text{var}|p_{rf}, p_{cnn}) = \frac{1}{1 + (\Lambda_{rf} \cdot \Lambda_{cnn})^{-1}}$$

$$\Lambda = \frac{P(p|\text{var})}{P(p|nv)}$$

This creates a single, highly robust unified prediction, mitigating the individual weaknesses of the RF and CNN models.

The Joint Bayesian Classifier



Probability density fits for the likelihoods $P_{\text{RF}}(p_i|\text{var})$ (left) and $P_{\text{RF}}(p_i|\text{nv})$ (right) derived from the RF model.

Performance Evaluation: Joint Bayesian Classifier

Testing set: 115 VAR and 625 NV.

Performance per class

Class	Precision	Recall	F1-Score
VAR	0,89	0,95	0,92
NV	0,99	0,98	0,98

Conclusion: The Joint Bayesian approach successfully maximizes the Recall (finding the rare variables) while maintaining an exceptionally high Precision (rejecting complex NV artifacts)

Overall performance

Model Architecture	Accuracy	F1-Score
Joint Bayesian Classifier	0,97	0,95

Pre-Filtering Strategy and Fields Sweeping

Field Sweeping: Pre-Filtering Strategy

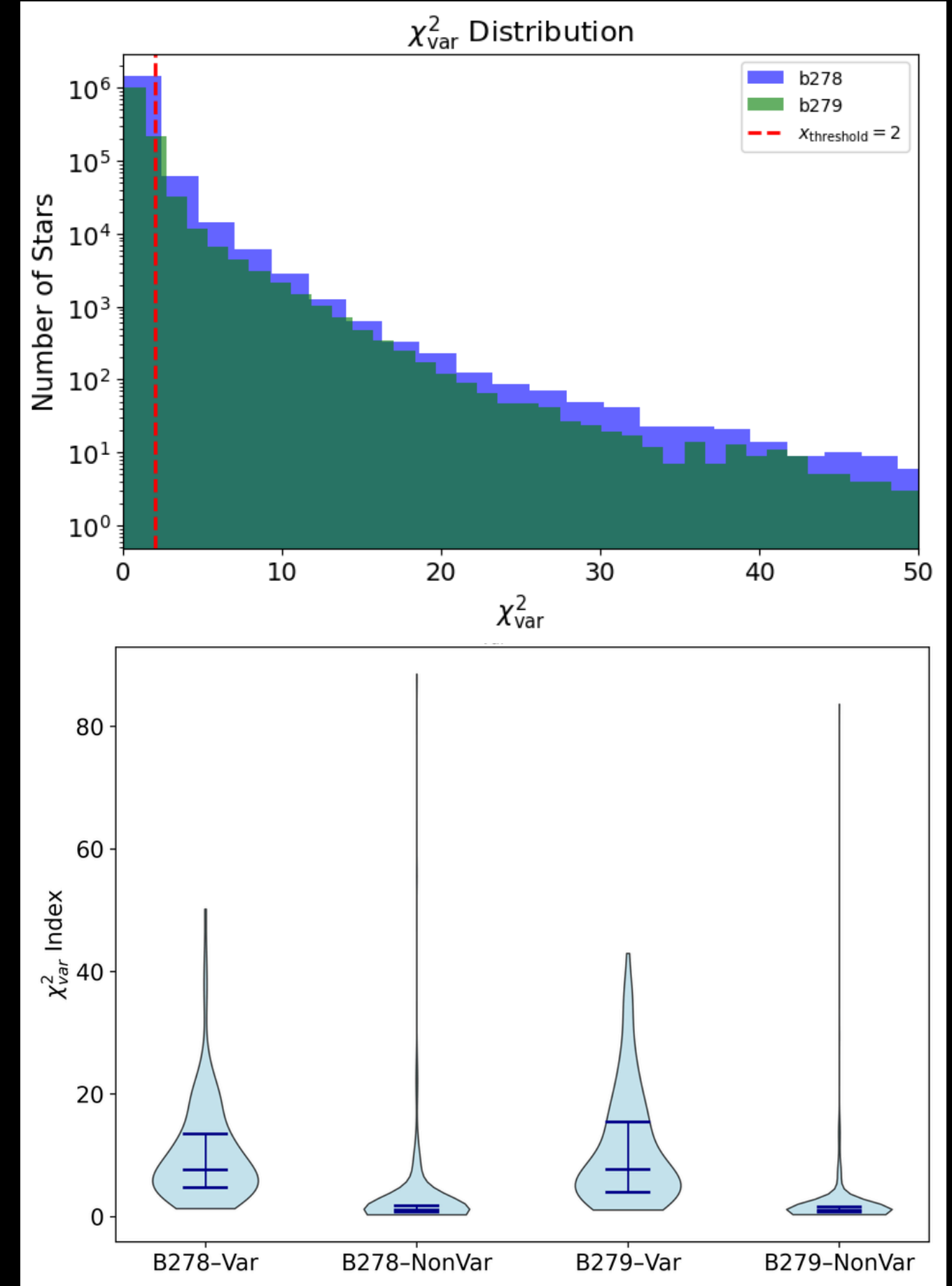
Computational Challenge

VVV fields (b278/b279) contain millions of stellar sources. Running complex ML models (like CNNs) across the entire dataset is computationally prohibitive and inefficient.

Empirical Threshold (χ^2_{var})

- A strict cut-off threshold was implemented based on the Variability Index (χ^2_{var}).
- **Purpose:** Rapidly and safely discard the vast majority of non-variable background (constant photometric noise).
- **Statistical Justification:** VAR populations exhibit heavy tails toward high χ^2_{var} values, making them clearly separable from the predominant Gaussian noise.

$$\chi^2_{\text{var}} = \frac{1}{N-1} \sum_{i=1}^N \frac{(m_i - \bar{m})^2}{\sigma_i^2}$$



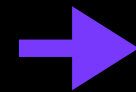
Visual Curation & Final Catalog

Despite the high precision of the Joint Bayesian model, the astrophysical complexity of the galactic bulge fields requires a final layer of human validation.

1. Model Candidates

The Bayesian Classifier outputs a list of sources with a probability $P(\text{var}) > 0.8$. This set is highly purified but still contains false positives from extreme anomalies not seen during training.

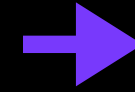
Bayesian Output



2. Visual Curation

Individual inspection of light curves and reference images to discard:

- **Blending:** Contamination from nearby bright stars.
- **Artifacts:** Stars on array edges (spurious high dispersion).



3. Curated Catalog

The end result is a robust catalog of genuine Variable Stars, ready for galactic studies.

Ready for Analysis

The model successfully filtered approximately 98% of the NV population while recovering about 95% of true VAR stars.

4. Multiclass Classification

Suggested variability classes are assigned based on the periods estimated through the minimum entropy method and the resulting light-curve morphology.

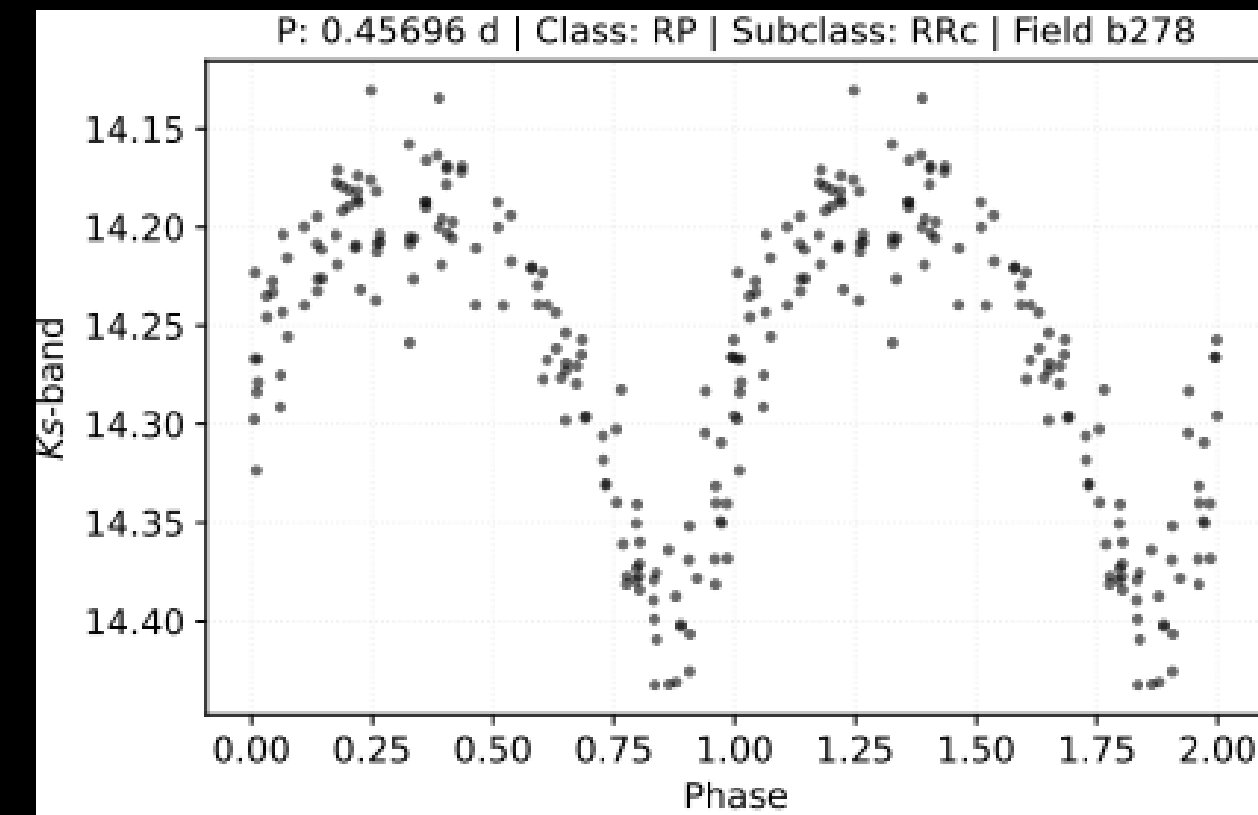
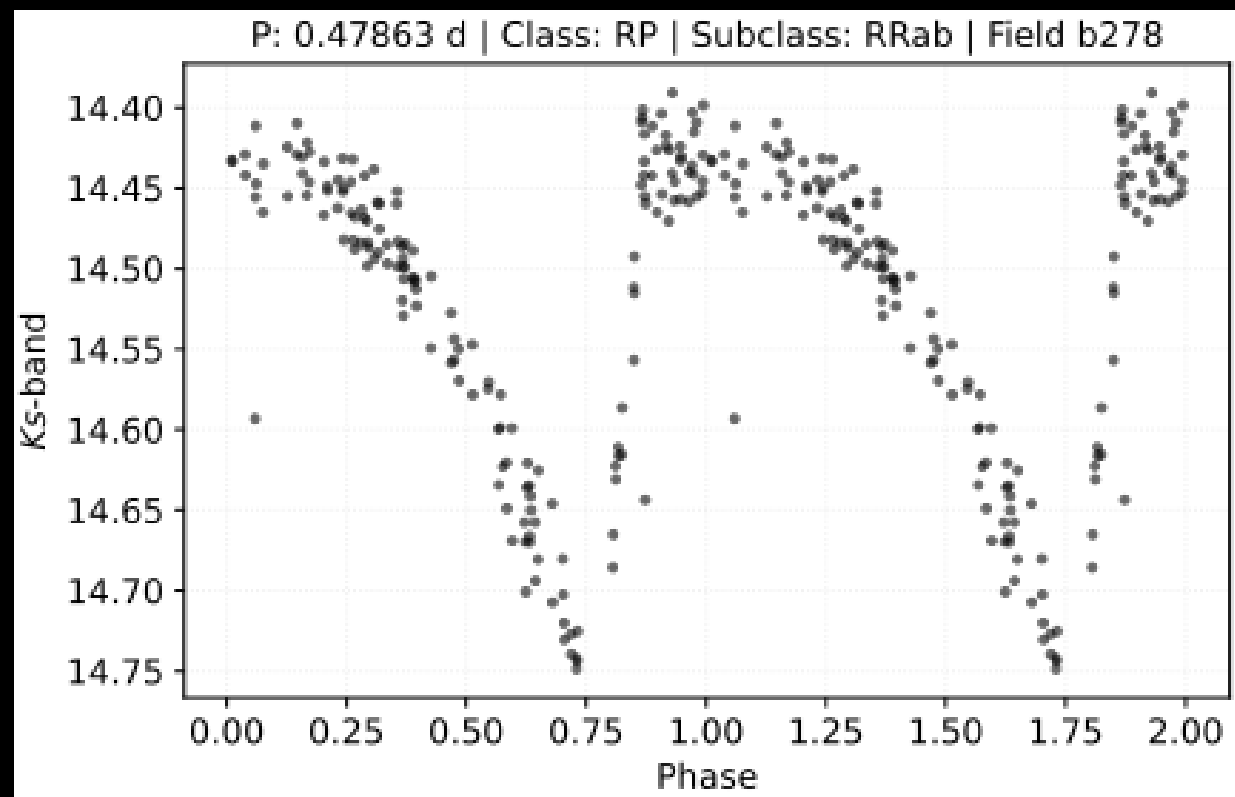
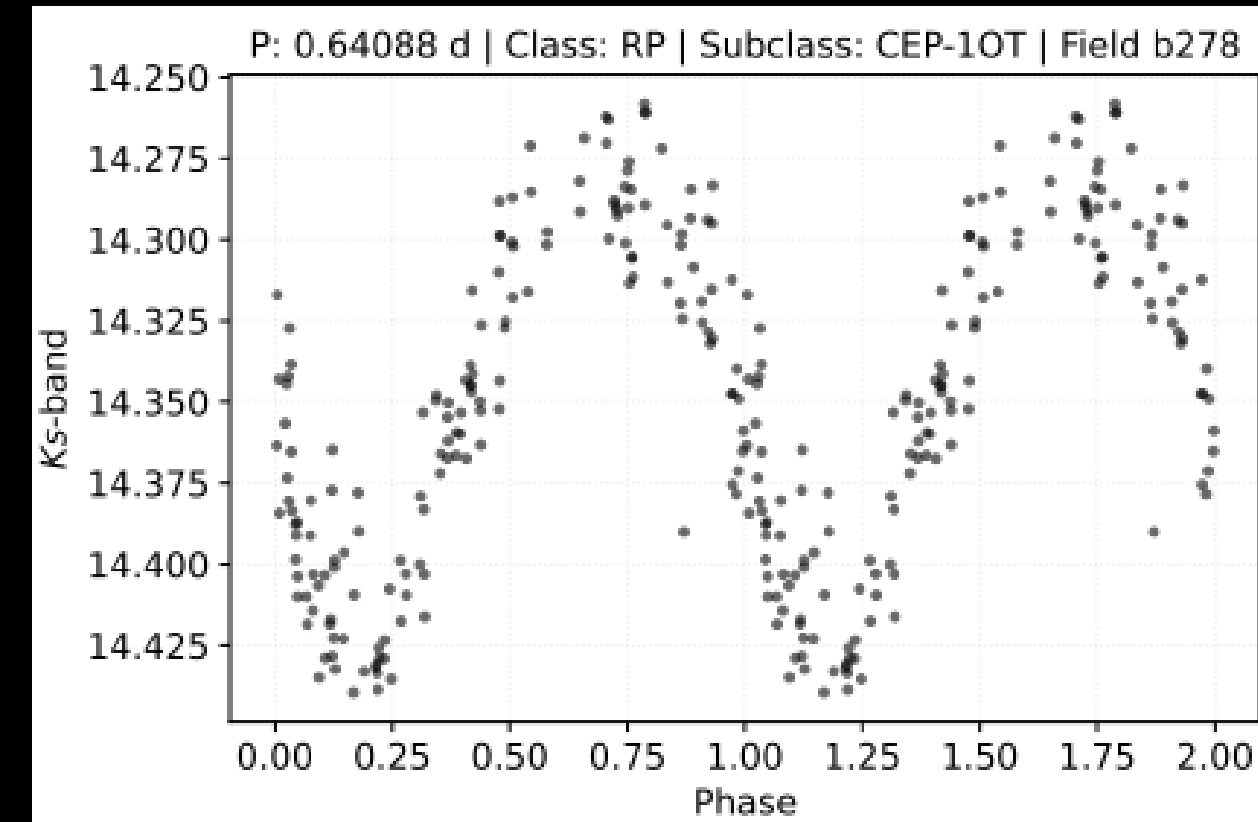
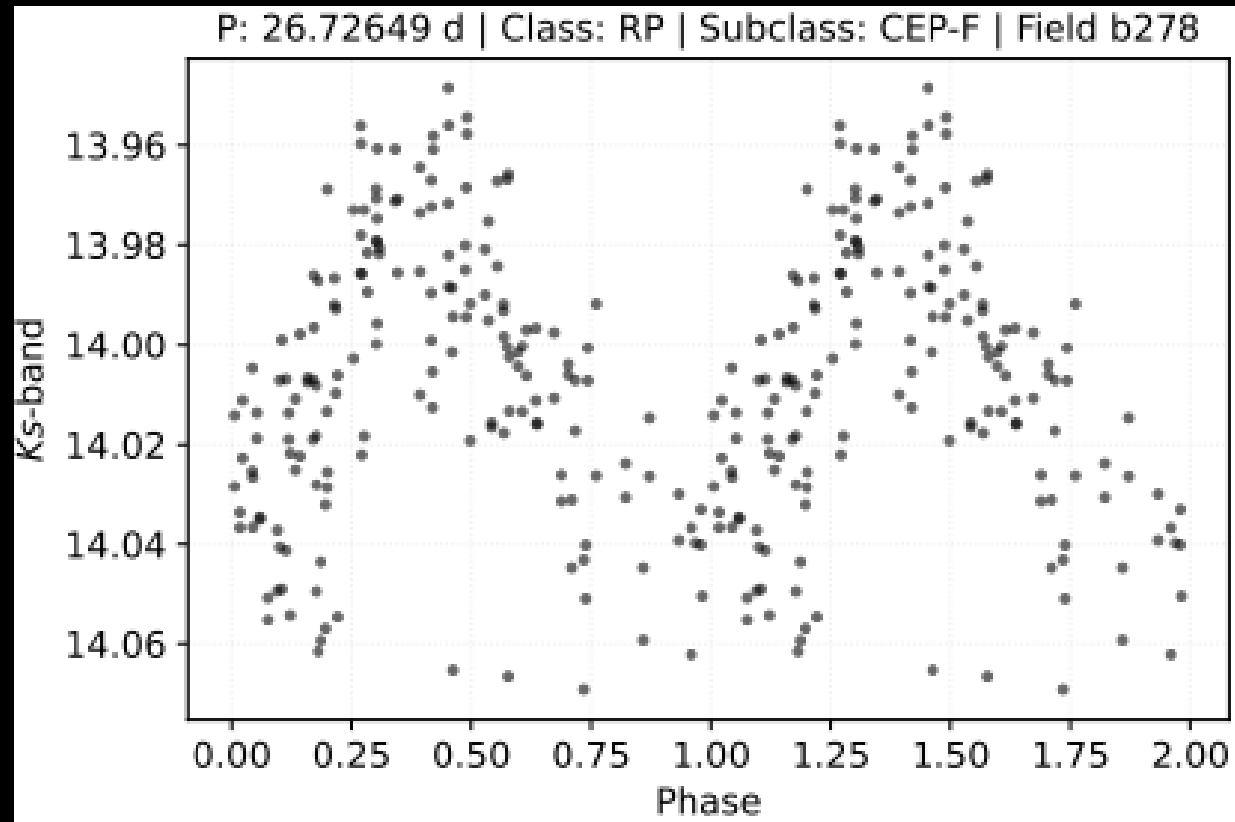
Hierarchical Statistical Classification

Core Strategy: A hierarchical automated pipeline designed to distinguish between Radial Pulsators (RP) and Eclipsing Binaries (E) based on light curve morphology.

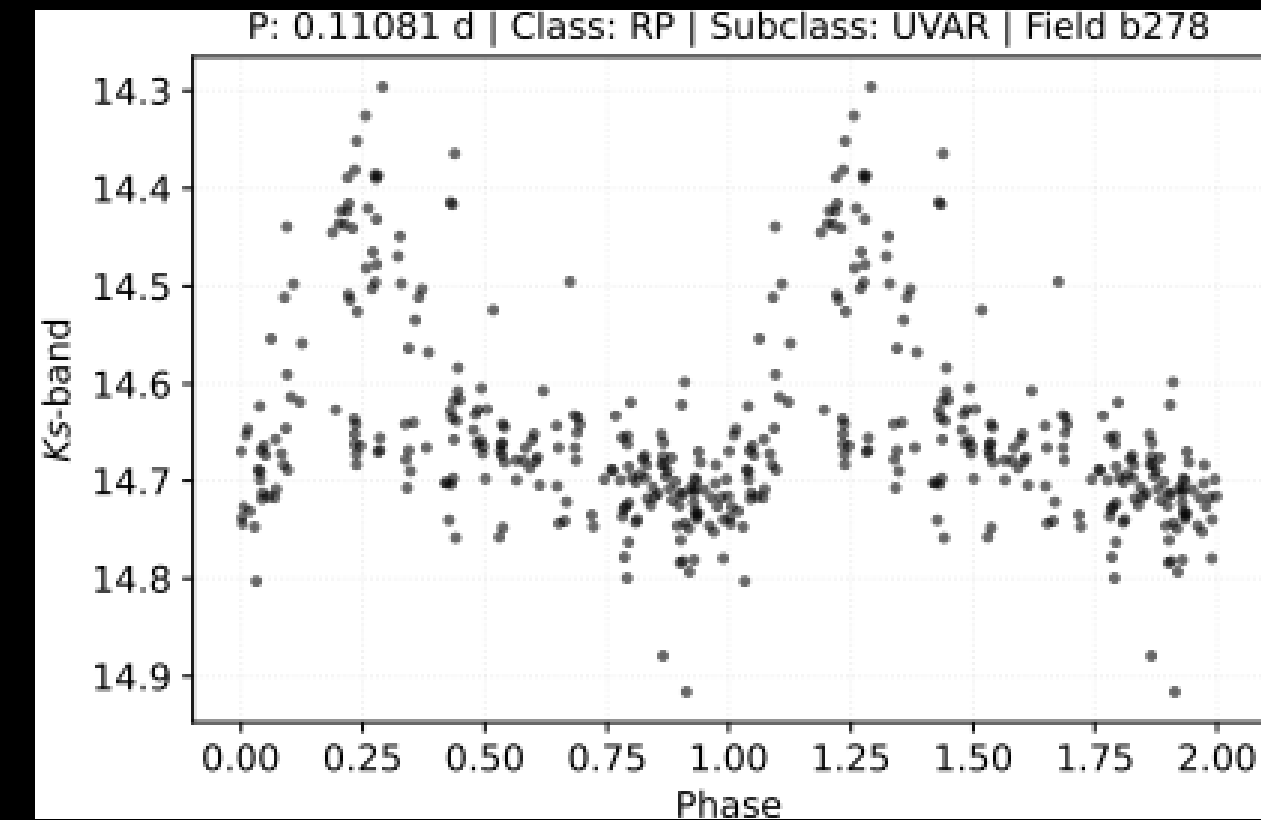
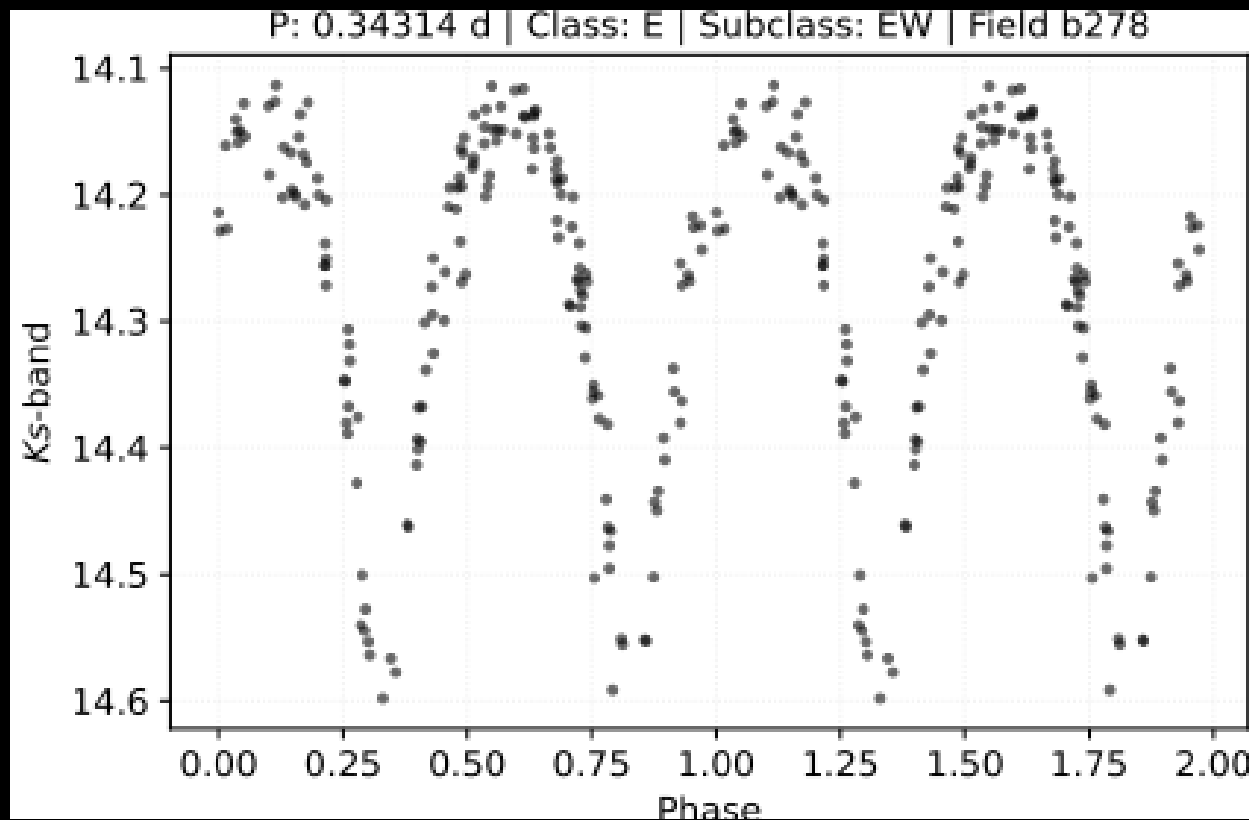
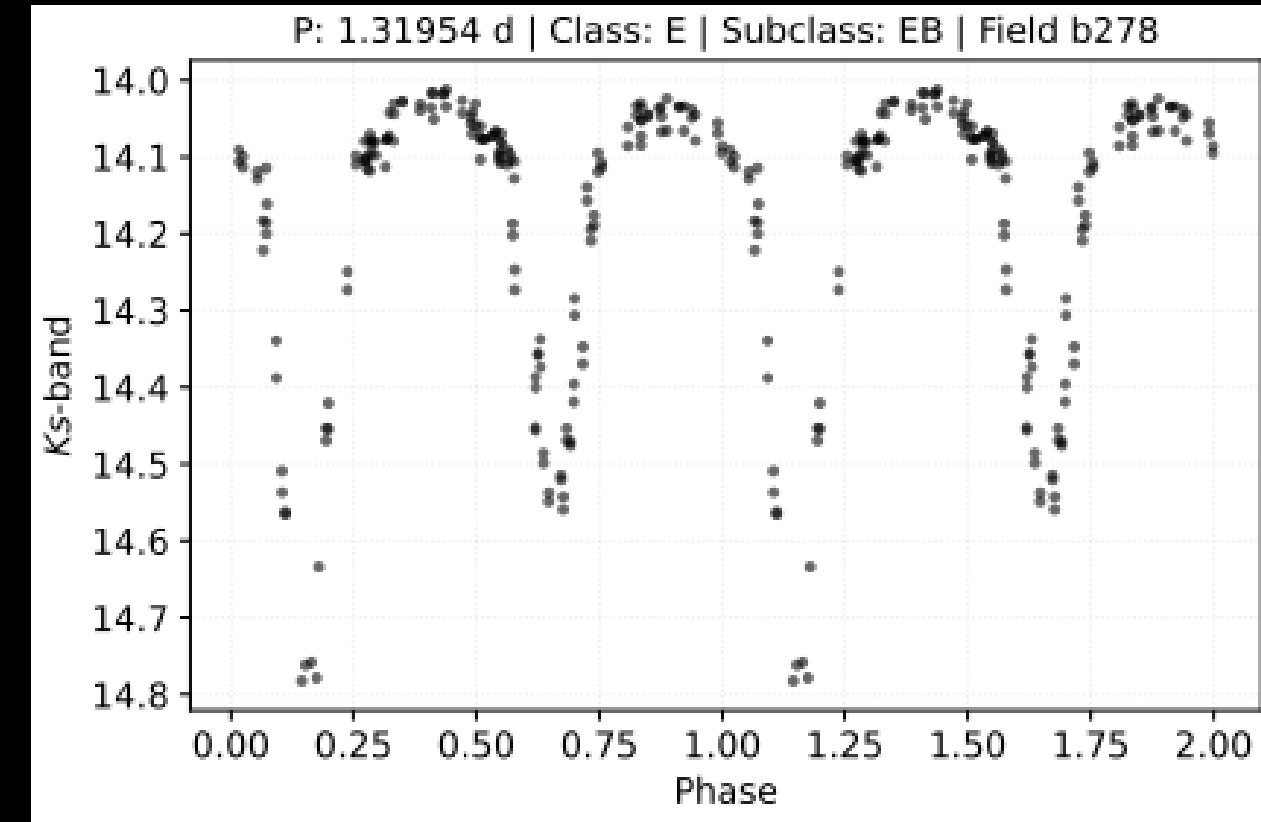
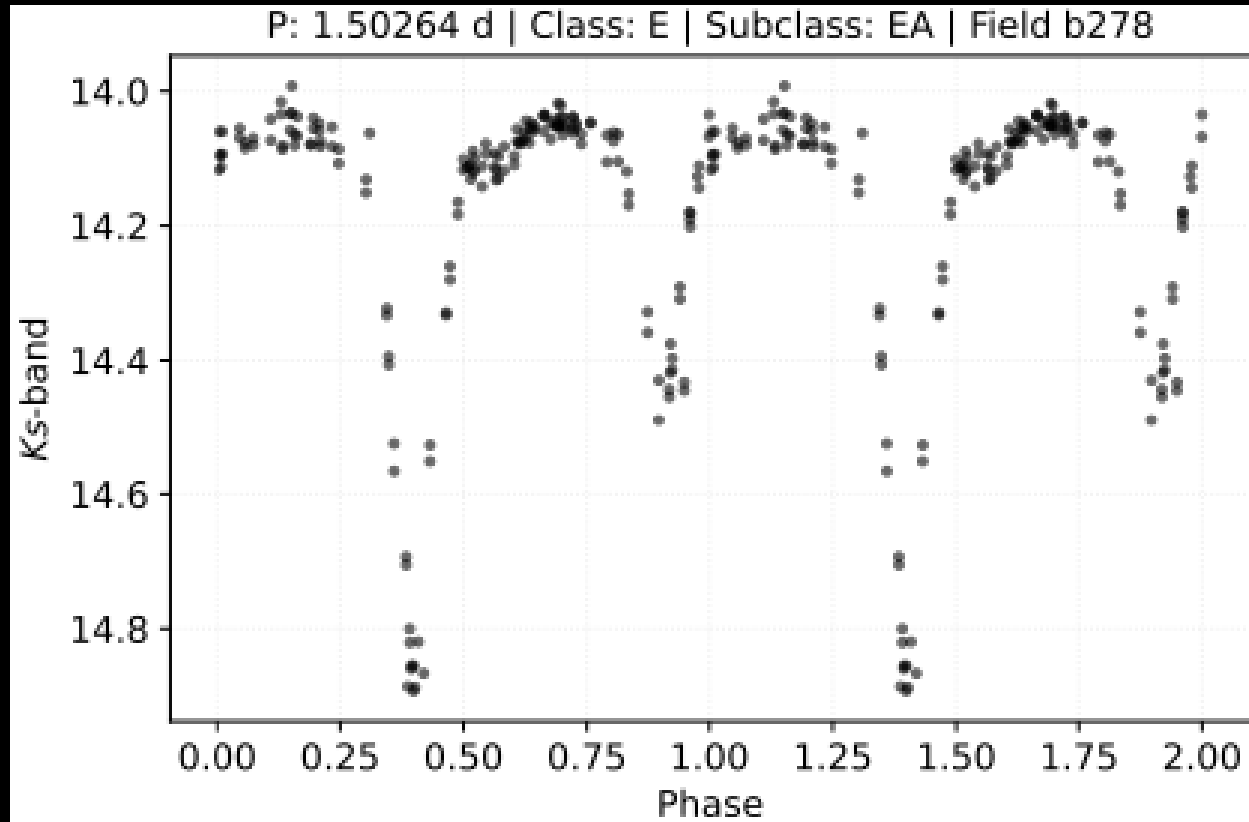
Subclass Refinement: Automated thresholds for Kurtosis, Flatness, and Depth Ratios to isolate specific types (RR Lyr, Cepheids, Algols, etc.).

Quality Control: High-noise or ambiguous sources are categorized as Unknown Variables (UVAR) to ensure catalog purity.

Main Class	Subclass	Classification Criteria / Ranges
Radial Pulsator (RP)	CEP-F	Period > 1.0 d; Symmetry < 0.45
	CEP-1OT	Period $\in [0.24, 10.0]$ d; Symmetry ≥ 0.45
	RRab	Period $\in [0.3, 1.0]$ d; Symmetry < 0.45
	RRc	Period $\in [0.2, 0.6]$ d; Symmetry $\in [0.45, 0.5]$
Eclipsing Binary (E)	EW	$R_{21} \in (0.1, 0.38)$; Kurtosis $\in (-2, 0)$; Flatness $\in (0.2, 0.6)$; Depth Ratio > 0.8; Amplitude > 0.3
	EA	Passes binary tests OR Kurtosis > 1.9; Flatness $\in (0.7, 1.2)$; Kurtosis $\in (1.9, 9)$
	EB	Passes binary tests; Flatness $\in [0.35, 0.7]$; Kurtosis $\in [0.0, 1.9]$
	RRc (Anti-filter)	Mimics EW but Amplitude < 0.3 and Kurtosis $\in (-2, 0.2)$; safely re-routed to RRc.
Unknown (UVAR)	UVAR	Fails to strictly meet the morphological boundaries of the aforementioned classes.

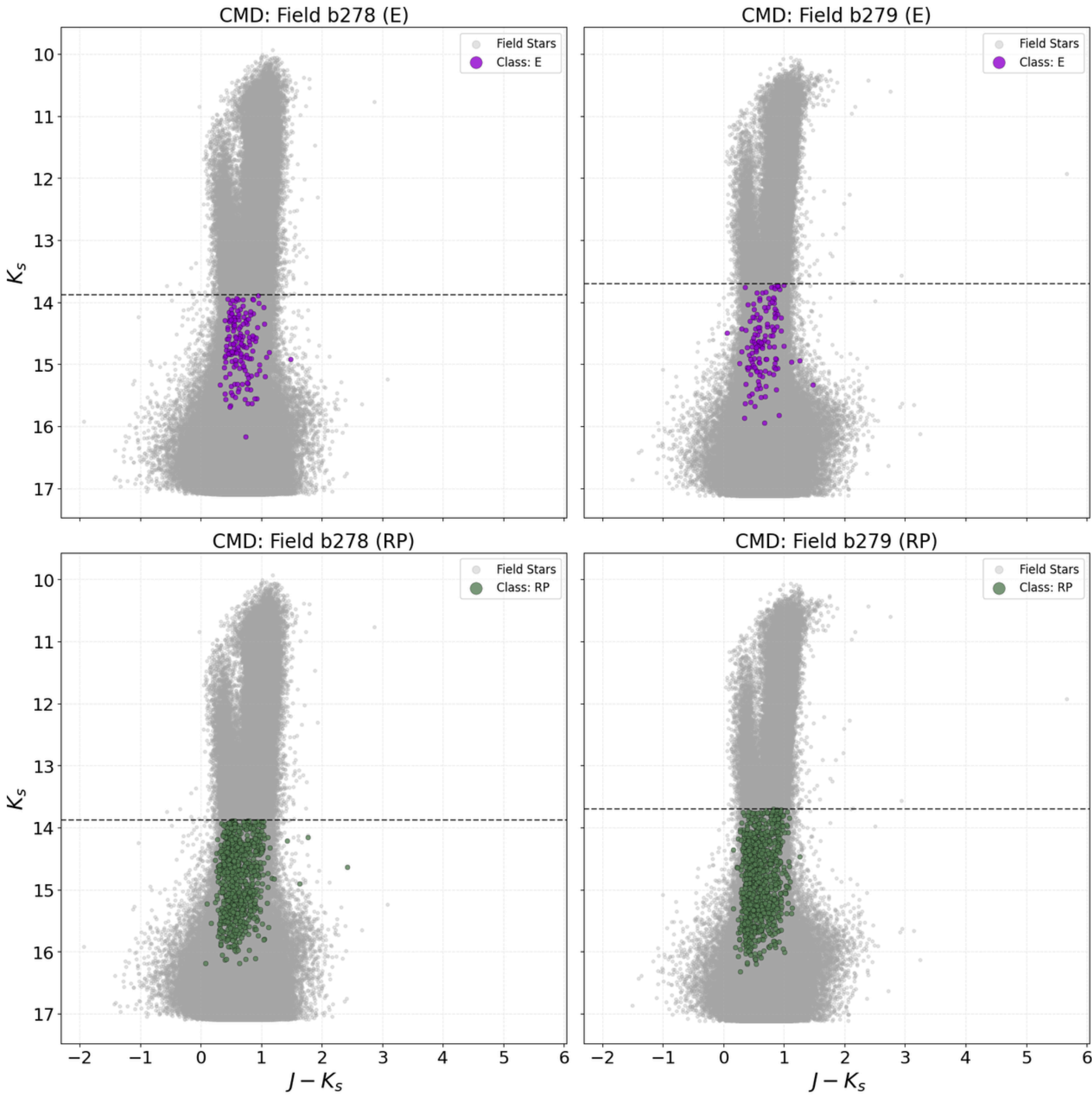


The complete repository of folded light curves and their respective classifications can be accessed at:
<https://github.com/DanielFajardo1/MINERVA-Light-Curves.git>



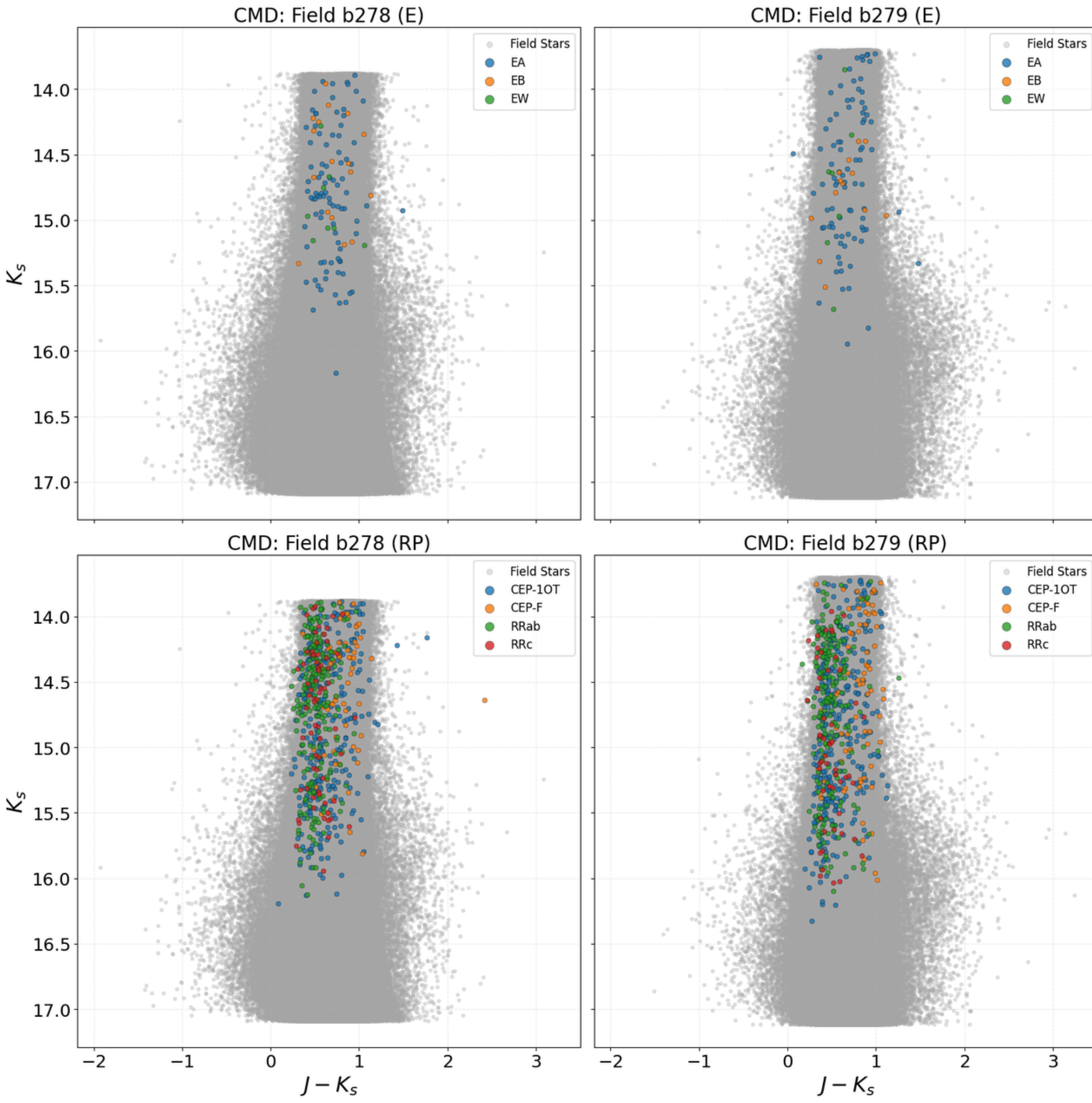
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Color-Magnitude Diagram



- **Vertical Morphology:** The broad vertical distribution is driven by severe **differential extinction**.
- **Saturation Regime:** Black dashed lines indicate the onset of saturation ($K_s < 13.88$ for b278; $K_s < 13.70$ for b279).
- **Morphological Contrast:**
 - Radial Pulsators (RP): Tightly concentrated due to their specific evolutionary stage within the **instability strip**.
 - Eclipsing Binaries (E): Highly dispersed, as they encompass stars from diverse masses and evolutionary stages.

Color-Magnitude Diagram



- **Consistency Check:** Subclass distribution aligns with expected physical trends despite the lack of extinction correction.
- **RP Temperature Gradient:** Clear hierarchy from reddest (coolest) to bluer (warmer) components:
 - CEP-F - CEP-10T - RRab - RRc
- **Binary Systems:** Majority are situated near the Main Sequence; outliers indicate systems with giant or subgiant components.

5. Minimum Entropy Variables (MINERVA) Catalog

A comprehensive, highly purified database of stellar variability.

The MINERVA Catalog: Results & Legacy

Final Catalog Yield

1,100

VAR (Field b278)

1,147

VAR (Field b279)

2,247

Total

Catalog Parameters per Source

- Unique Source Identifiers (IDs).
- Precise Equatorial Coordinates (RA/Dec).
- Multiband NIR Photometry (*J*, *H*, and *Ks* bands).
- High-precision variability periods derived via the entropy framework.
- Suggested variability classifications for the high-confidence subset.

Galactic Structure

Galactic Evolution

Distance Scale

Overcoming the Contamination Barrier

Previous efforts, notably the VIVA-I catalog (Ferreira Lopes et al., 2020), identified ~44.9 million candidates. However, it suffers from a massive 10:1 contamination ratio, diluting the true variable population ($\sim 10^6$) amidst stochastic noise and limiting its direct applicability.

- **The MINERVA Advantage:** Our strict statistical selection (χ^2_{var}) combined with the Joint Bayesian validation strategy guarantees significantly **higher sample purity**.
- **Survey-Wide Projection:** Extrapolating this pipeline across all 348 VVV tiles projects a yield of $\sim 10^5$ variable stars.

The VVV Framework Legacy

Scaling this pipeline establishes MINERVA as the groundwork for the most extensive and reliable catalog of high-confidence variable stars within the VISTA project to date.

The MINERVA Catalog: Results & Legacy

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Galactic Structure

Galactic Evolution

Distance Scale

Variability Class	Field b278	Field b279	Total
Cepheids	367	419	786
CEP-F	63	77	140
CEP-1OT	304	342	646
RR Lyrae	301	272	573
RRab	233	219	452
RRc	68	53	121
Eclipsing Binaries	121	113	234
EW	8	7	15
EA	96	93	189
EB	17	13	30
UVAR	311	343	654
Total Subset	1,100	1,147	2,247

RR Lyrae Surface Density: MINERVA vs. Literature

MINERVA:

- 573 RRab stars detected
- **Survey area:** 3.3 deg^2 (two $1.5^\circ \times 1.1^\circ$ fields)
- **Surface density:** 174 RRab/deg^2
- **Location:** Inner Galactic bulge, maximum density region

Contreras Ramos et al. (2018):

- 960 RRab stars detected
- **Survey area:** 8.73 deg^2 (100 arcmin radius from GC)
- **Surface density:** 110 RRab/deg^2
- **Coverage:** Broader Galactic Center region

- MINERVA density $1.58\times$ higher than Contreras Ramos et al. (2018)
- Fully consistent with expected density gradient
- Both studies use VVV/VVVX PSF photometry (VIRAC2)
- Confirms our fields probe peak density of bulge RRL population

Conclusions

1. Developed and implemented an integrated framework to classify stellar variability in the highly obscured Galactic bulge.
2. Applied the minimum entropy method to overcome irregular sampling limitations, extracting highly discriminative features with strong statistical weight for classification.
3. Produced a highly pure catalog (MINERVA) of 2,247 confirmed variable stars (1,100 in field b278; 1,147 in b279).
4. Extrapolating this framework to the entire VVV survey establishes MINERVA's potential to become the largest and most reliable high-purity catalog of confirmed near-infrared variable stars to date.

Future Work:

1. **Pipeline Optimization and Refinement:** Fine-tuning classification parameters (entropy cell size, period grid windows, and Bayesian thresholds) to minimize model complexity while maintaining high recovery rates.
2. **Characterization of the Saturated Regime and Reference Standards:**
 - a. **Saturated Regime Exploration:** Investigating the impact of saturation on stellar populations in fields b278 and b279 to establish a baseline for future near-infrared studies.
 - b. **Standard Star Candidates:** Identifying and characterizing non-variable (NV) stars to provide the astronomical community with much-needed photometric calibrators in the infrared.
3. **Expanding the Parameter Space:** Investigating variability outside the current scope, including ultra-short periods (Delta Scuti stars, $p < 15$ min) and long-period variables (Miras, $p > 95.5$ days).
4. **Strategic Survey Expansion:** Scaling the MINERVA pipeline to high-density Galactic bulge fields (e.g., 293–296 and 307–310). Note: These areas offer high cadence and stellar density, promising a significant increase in the confirmed variable population.
5. **Cross-Match & Comparative Analysis:**
 - Systematically cross-identify MINERVA variables with literature catalogs
 - Compare properties (periods, amplitudes, metallicities) with benchmarks:
 - RR Lyrae: Contreras Ramos., et al. (2018), Elorrieta., et al. (2016), Alonso-García., et al. (2025)
 - Other variables: VIVACE (Molnar., et al., 2022), VIVA-I (Ferreira Lopes., et al., 2020)

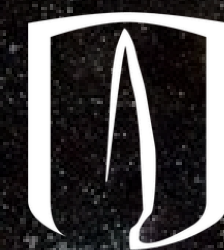
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THANK YOU

Questions & Discussion

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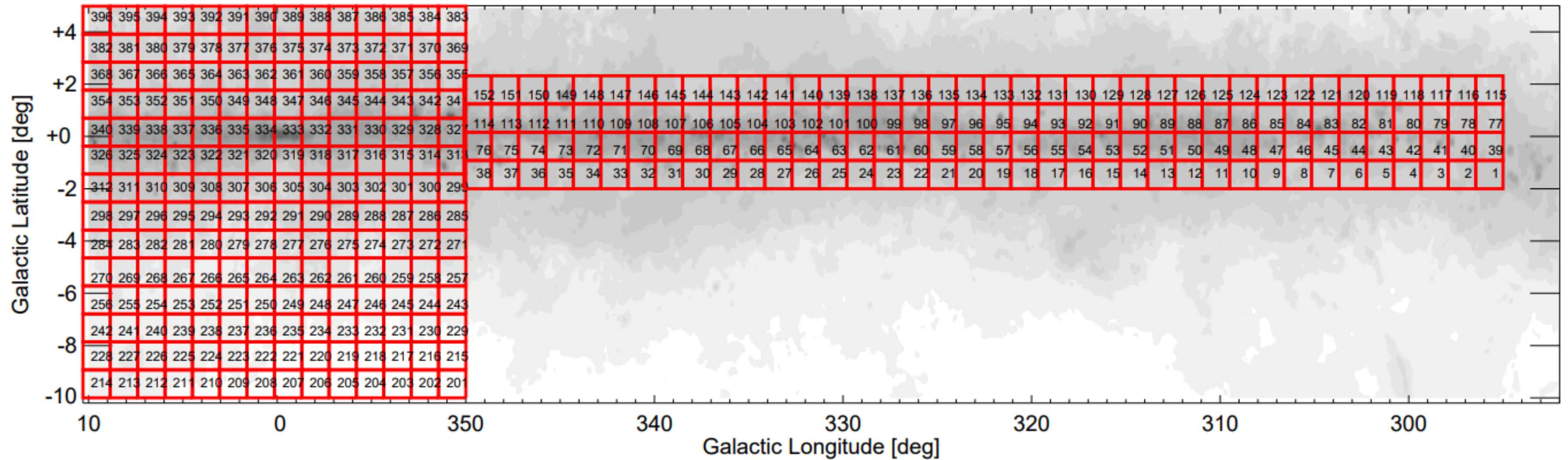


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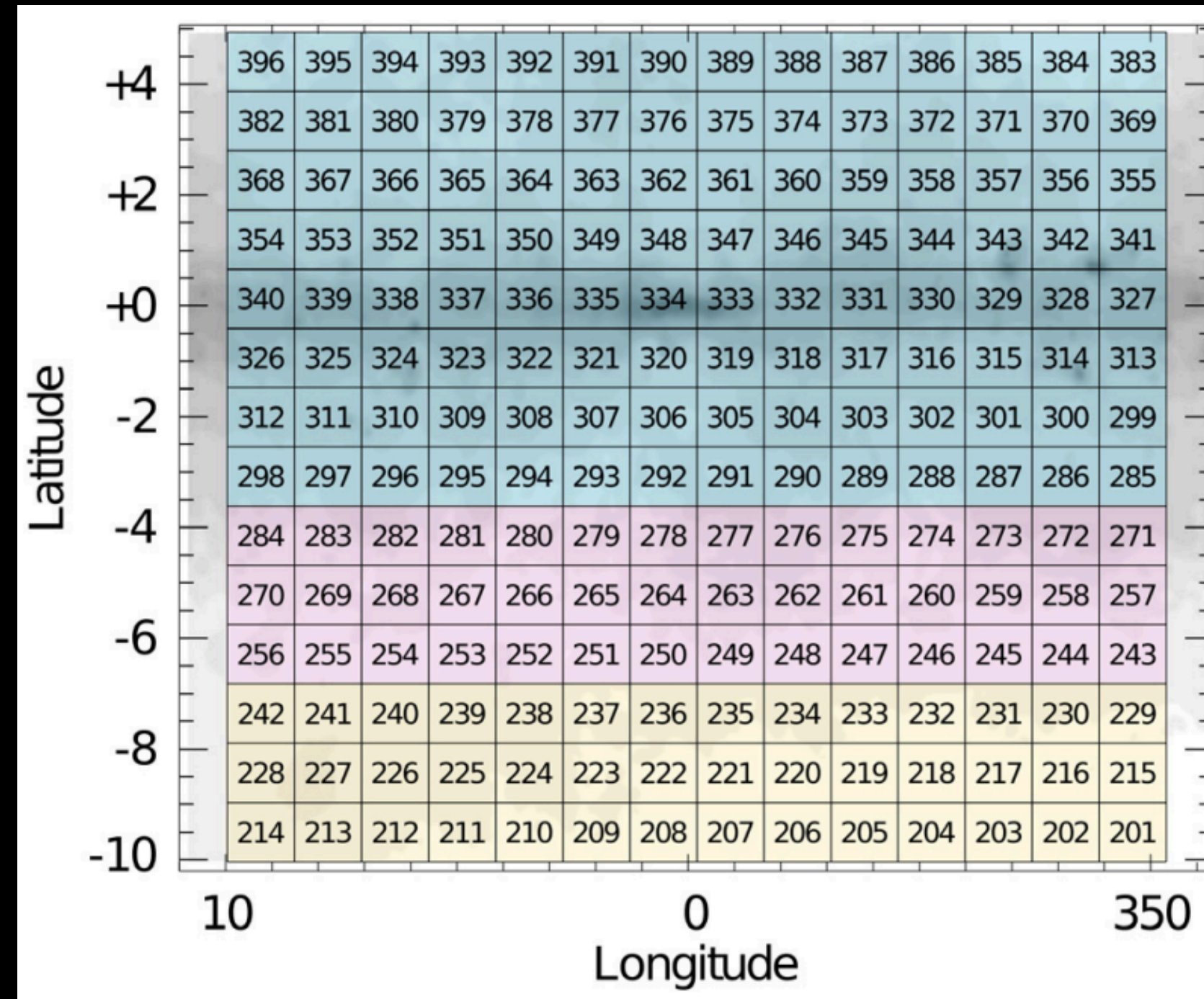
Supplementary Material

The VVV Survey



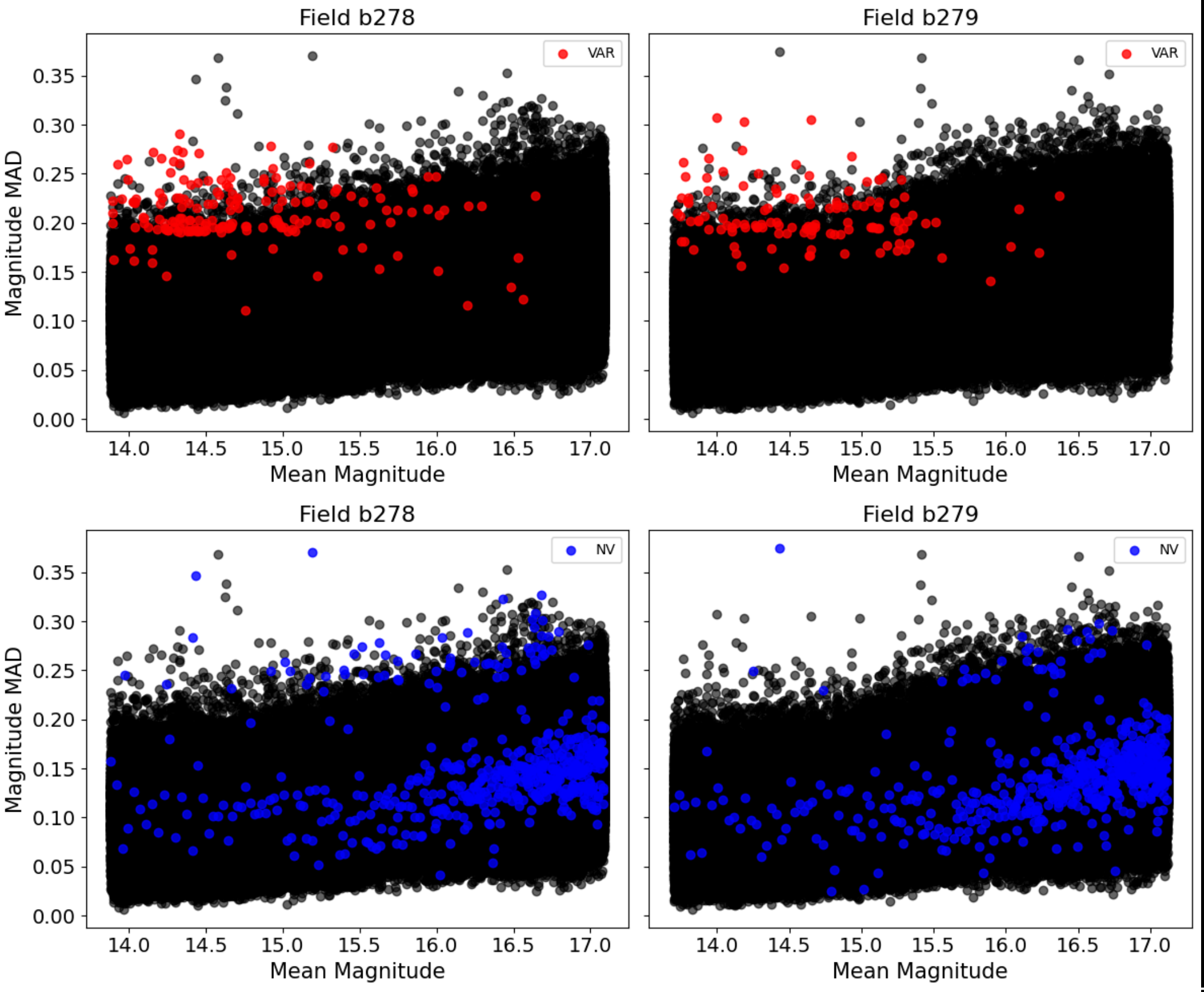
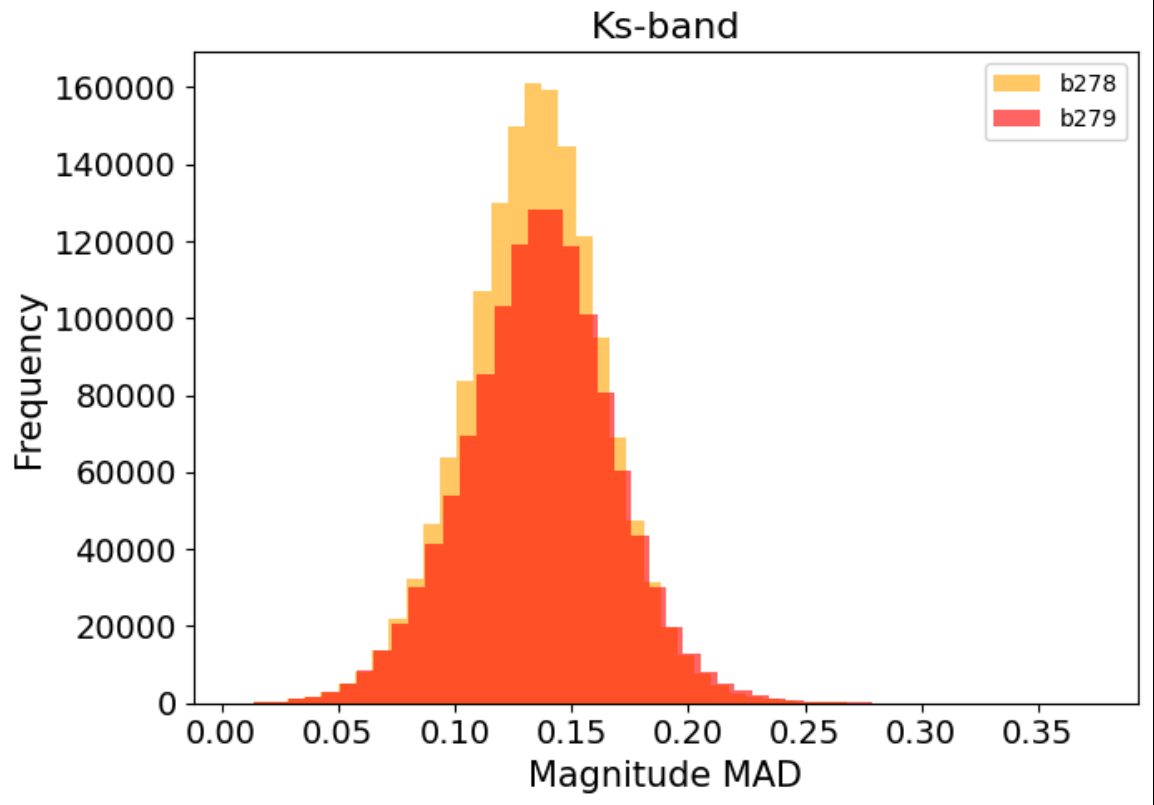
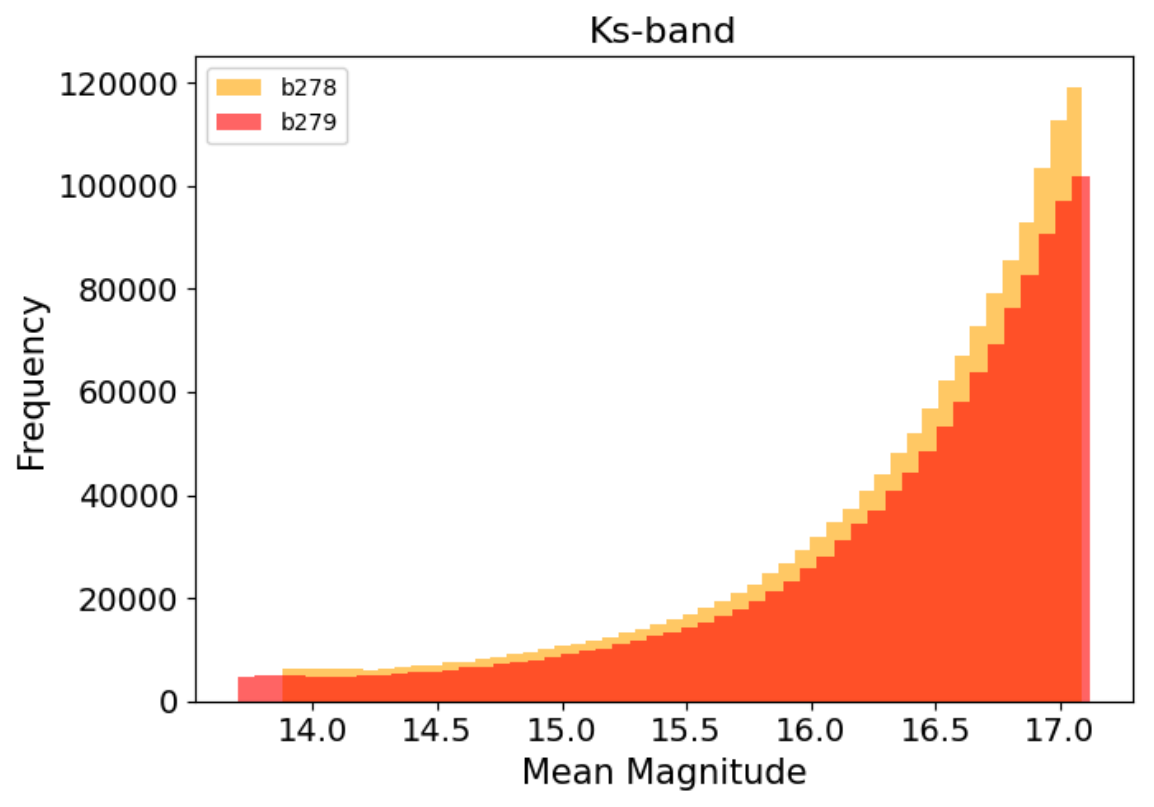
Taken from Saito et al. (2012)

Galactic Bulge



Taken from Gonzalez et al. (2011)

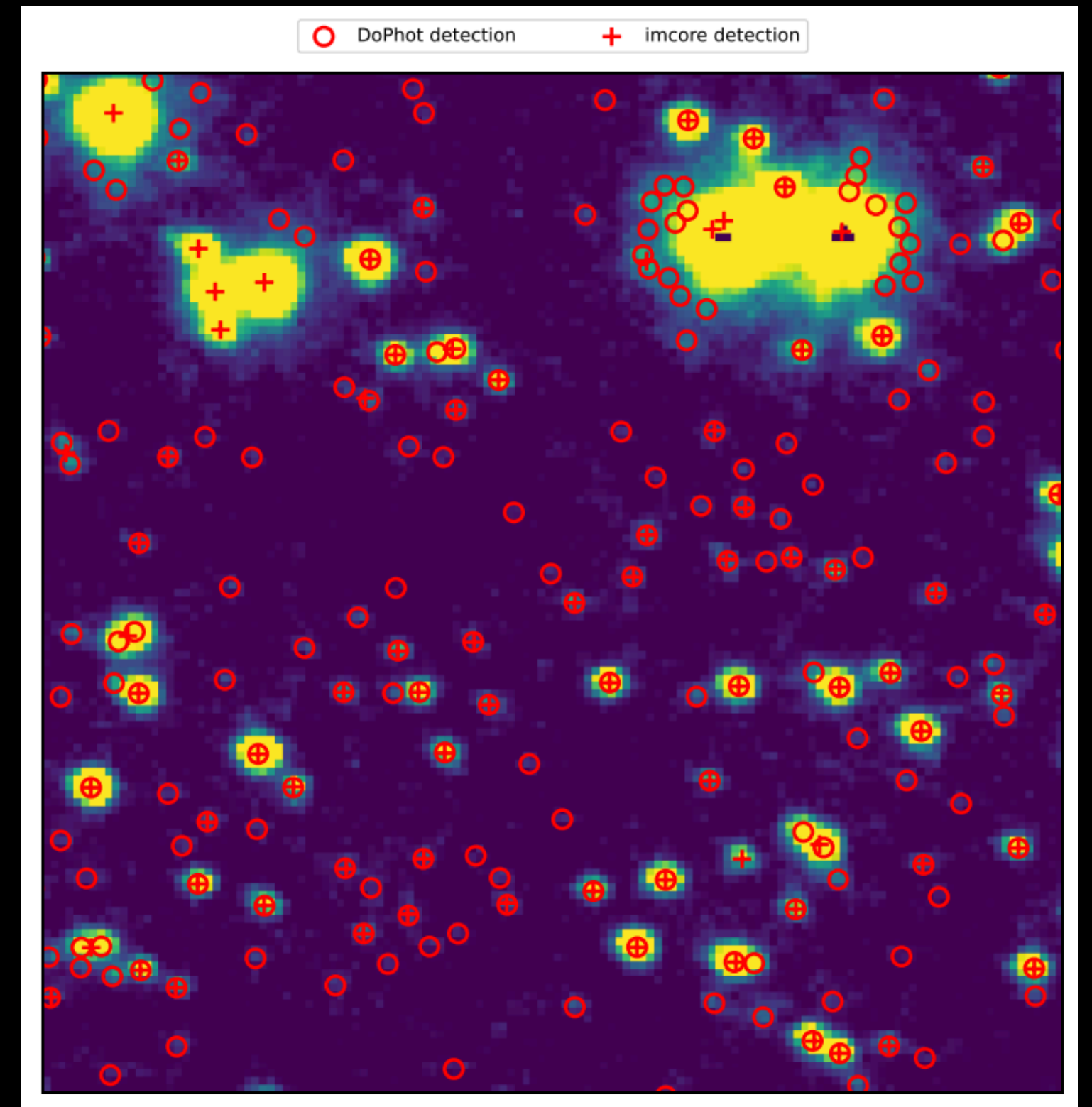
Ground Truth Dataset



VVX_VIRAC2_time_series

- **Survey (VVV → VVX):** VVV (2009–2015) mapped 560 deg² establishing ~100 epochs in the Ks-band. VVX (2016–2023) expanded to 1,700 deg² and doubled the temporal baseline (>100 Ks-band epochs).
- **Photometry (VIRAC1 → VIRAC2):** VIRAC1 utilized aperture photometry. VIRAC2 implemented PSF (Point Spread Function) fitting via DoPhot.

Comparison of CASU IMCORE and DOPHOT detections in a sample image cutout. DOPHOT (red circles) reaches fainter limits and deblends sources more effectively, but produces more spurious detections near bright stars and **misses some saturated sources**. IMCORE (red crosses) more reliably recovers real and saturated stars, though with poorer deblending and a brighter detection limit.



Taken from Smith, Leigh C., et al. (2025)

VISTA Observational Strategy

- **Flexible Cadence:** Temporal separation is highly variable. Integration time is 162 s per tile, yielding a mapping rate of $\sim 30 \text{ deg}^2/\text{hour}$.
- **Heterogeneous Epoch Density:**
 - New VVVX Extensions: 23–50 epochs.
 - Inner Galactic Bulge: ~ 100 additional epochs.
 - Baade's Window: Up to 352 total combined epochs (VVV + VVVX), representing maximum density.

Temporal Requirements (Saturated Stars)

- **Minimum Baseline:** Saturated targets dependent on specialized PSF photometry require 50–100 epochs over several years to properly characterize variability and suppress aliasing.
- **Amplitude Sensitivity:** Detection limits for variables (e.g., RR Lyrae) scale strictly with data points:
 - 80 epochs (~ 3 years): Detects amplitudes down to $A_K=0.05\text{--}0.07\text{mag}$
 - 180 epochs: Threshold improves to $A_K=0.03\text{--}0.05\text{mag}$.

Transients and Rare Stellar Populations

Transient Events (Novae & SNe)

- **Highly Active Novae Region:** The Bulge presents a high classical nova rate (13.8 ± 2.6 per year measured by OGLE).
- **Survey Synergy:** High-cadence surveys (ASAS-SN) detect fast novae, while IR surveys (Palomar Gattini-IR, Gaia) uncover heavily reddened/obscured events.
- **Supernovae Dynamics:** Core-collapse SNe are extremely rare due to the lack of massive young stars. However, **Thermonuclear SNe (Type Ia)** originating from ancient binary systems do occur.

Scarcity of True Be Stars

- **Astrophysical Constraints:** Be stars are rapidly rotating, massive main-sequence B-type stars (typically 3–15 $M_{\odot}$).
- **Evolutionary Improbability:** Because the Bulge's dominant population is ancient (>8 Gyr), any primordial massive B-type stars have long since evolved off the main sequence or exploded.
- **Foreground Contamination:** Any Be stars detected along the line of sight are highly likely to belong to the younger overlapping inner Galactic disk, not the intrinsic Bulge population.

Galactic Bulge: Age & Metallicity Profile

Stellar Age Distribution

- **Dominant Ancient Population:** The Bulge is primarily composed of old stars (>8 Gyr). The central region mirrors metal-rich globular clusters (~ 13 Gyr).
- **Younger Sub-Components:** A non-negligible fraction of younger stars ($\sim 2\text{--}5$ Gyr) exists, predominantly at super-solar metallicities.
- **Metallicity-Age Relation:** Metal-poor stars are strictly ancient, while metal-rich populations exhibit a much broader age range (up to 35% are <8 Gyr).

Chemical Evolution ($[\text{Fe}/\text{H}]$)

- **Bimodal Distribution:** The Bulge exhibits a broad metallicity range ($-1.5 \lesssim [\text{Fe}/\text{H}] \lesssim +0.5$). Peaking distinctly at $[\text{Fe}/\text{H}] \sim -0.6$ (metal-poor) and $[\text{Fe}/\text{H}] \sim +0.3$ (metal-rich).
- **Modern Consensus:** Represents a complex mix of a rapidly-formed ancient population and secularly-evolved inner-disk stars.

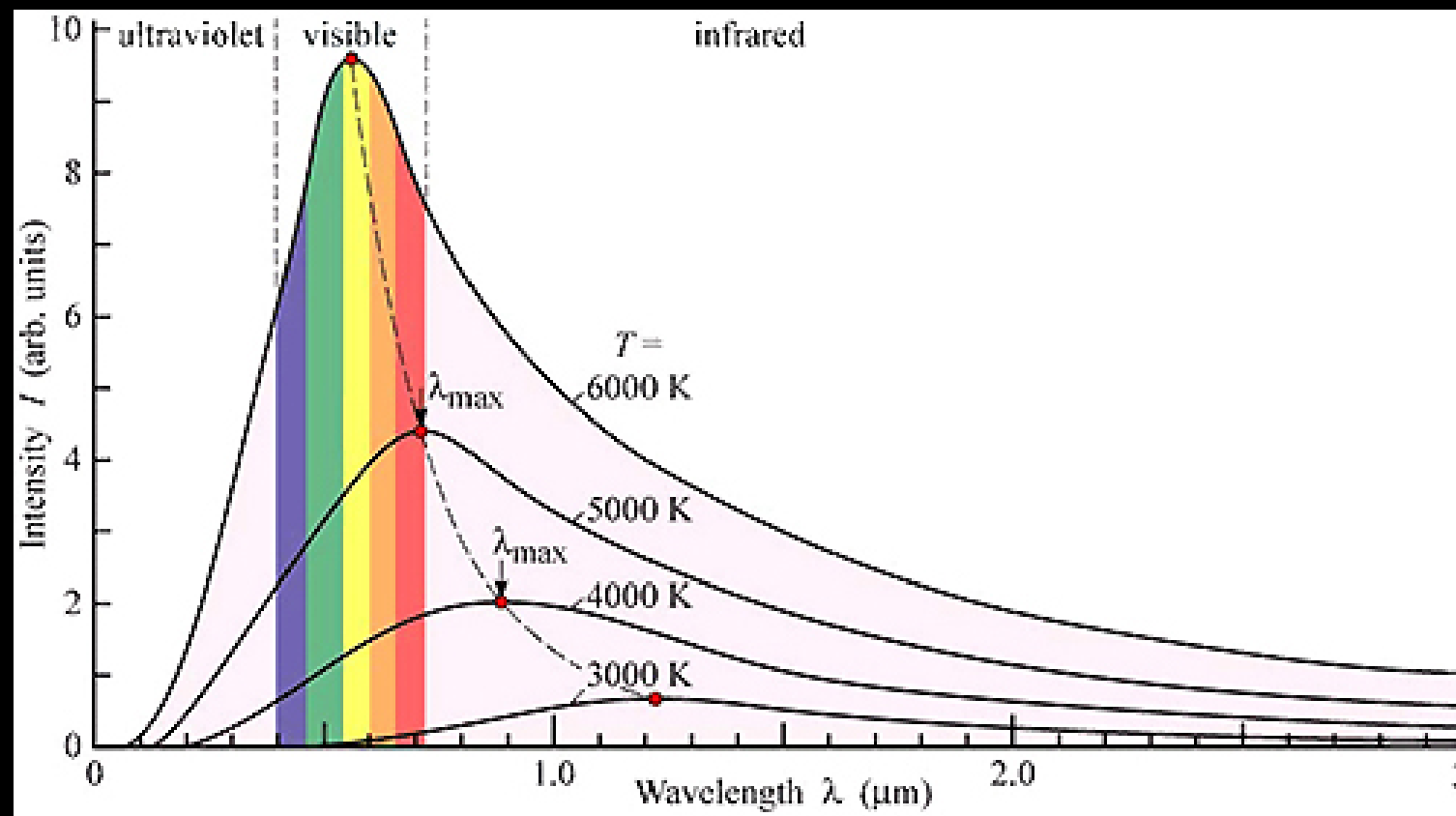
Amplitude Attenuation: NIR vs. Optical Regimes

Radiative Physics (Planck's Law)

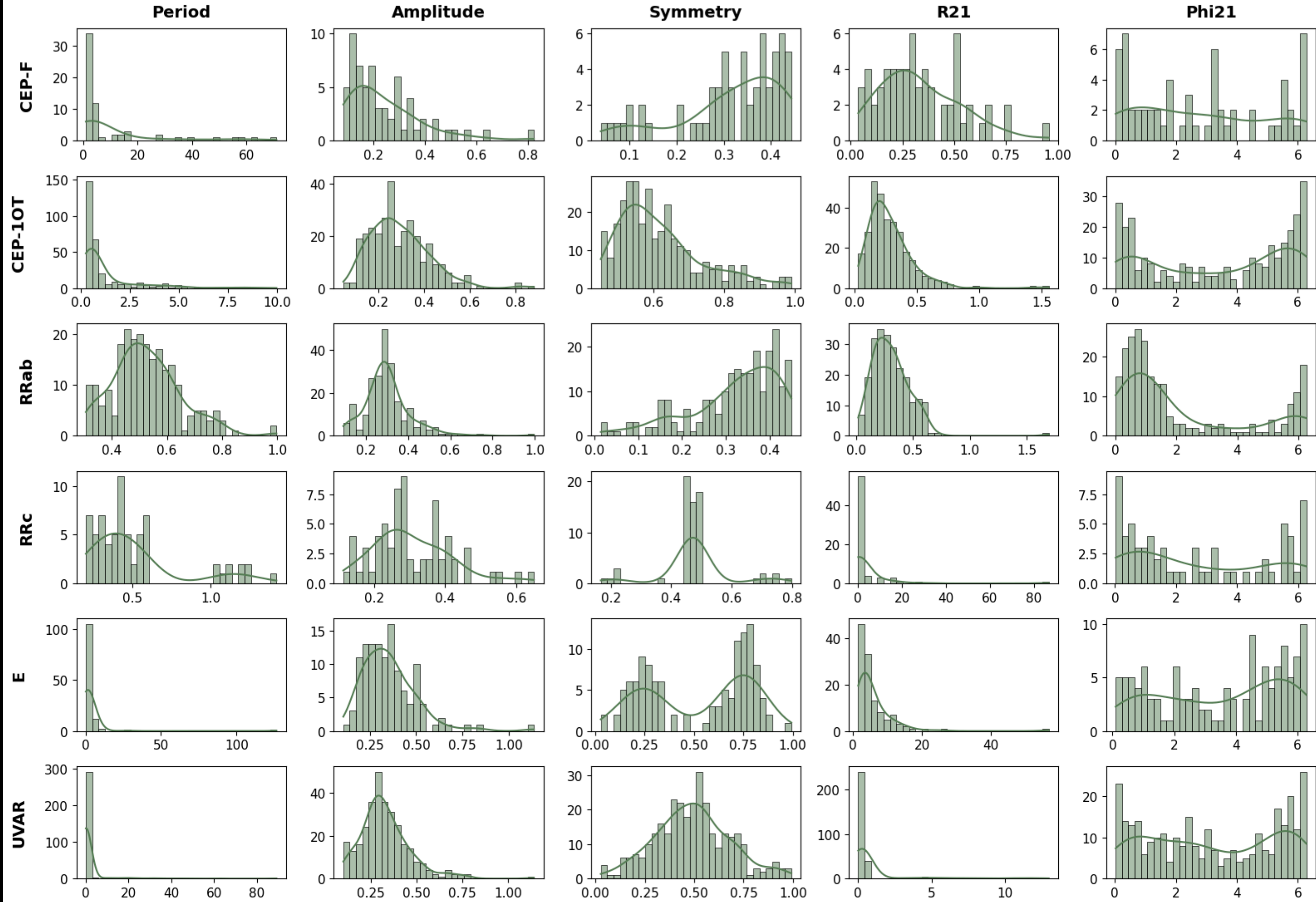
- **Reduced Temperature Sensitivity:** At longer wavelengths (NIR, 1–3 μm), the Planck function exhibits significantly less sensitivity to effective temperature fluctuations (ΔT).
- **Damped Response:** For a given ΔT during the pulsation cycle, the resulting variation in NIR emission is physically much smaller than in the optical band.

Photospheric Depth and Kinematics

- **Photospheric Depth Dependence:** NIR observations probe deeper and hotter layers of the stellar photosphere, where the temperature gradient is shallower and relative temperature fluctuations are intrinsically reduced compared to the optical regime.
- **Impact on Variability Amplitudes:** Because flux variations become less sensitive to temperature changes at longer wavelengths, light-curve amplitudes systematically decrease from the optical to the NIR bands.



Variability Subclass Characterization | Field b278



Variability Subclass Characterization | Field b279

